The Short-Run Consumer Surplus of Uber: Evidence from Four U.S. Cities

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Abstract

Estimating consumer surplus is challenging because it requires identification of the entire demand curve. We exploit two key features of Uber's data to overcome these difficulties: (1) sharp discontinuities in Uber's real time "surge" pricing, and (2) potential customers signaling they are in the market by opening the app before they know current prices, which appears to eliminate an important source of bias in estimating demand. We are able to estimate demand elasticities at numerous points along the demand curve, and we then use these elasticity estimates to estimate consumer surplus. Using almost 50 million individual-level observations, we estimate that in 2015 the UberX service generated about $2.6 billion in consumer surplus in the four U.S. cities included in our analysis. For each dollar consumers spent on UberX rides, about $1.40 of consumer surplus is generated. Back-of-the-envelope calculations suggest that the overall consumer surplus generated by the UberX service in the United States in 2015 was $6.0 billion. Reweighting high-surge sessions to match the observable characteristics of base-price riders raises the four-city estimate to about $3.2 billion, and the alternative assumptions we examine yield a range of roughly $2.4 billion to $3.4 billion. These are short-run estimates that hold all competing transportation options fixed.

Keywords: consumer surplus, demand estimation, discontinuity design, Uber.
JEL: D1, D2, D4, L1.

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Introduction

For over 250 years, economists have recognized the importance of consumer surplus when making welfare calculations.[2] Consumer surplus is a critical input into many economic policies, such as antitrust analysis, the valuation of non-market goods, and measuring the value of innovation (e.g., Williamson 1968, Willig 1976, Bresnahan 1986).

In practice, however, obtaining convincing empirical estimates of consumer surplus has proven to be extremely challenging. We typically observe only one equilibrium point that balances supply and demand. Variations in that equilibrium across time and space are generally the result of a combination of supply-driven and demand-driven shocks and thus are of little use in this regard. Second, computing consumer surplus requires not just local elasticities in the neighborhood of the equilibrium, but rather knowledge of the entire demand curve. Usually the observed price variation spans only a fraction of the demand curve, making strong functional form assumptions (e.g., iso-elastic demand) necessary to produce consumer surplus estimates.

Given the centrality of demand to the field of economics, it is unsurprising that an enormous body of economic research focuses on demand estimation (see, for instance, Deaton 1986). The key to estimating demand elasticities is to isolate shifts in the supply curve that are orthogonal to shifts in the demand curve. In recent years, a great deal of work has focused on the development of new techniques for generating demand and welfare estimates in differentiated product markets. (Baker and Bresnahan 1988, Berry et al. 1995, Nevo 2000, Petrin 2002, Mortimer 2007, Crawford and Yurukoglu 2012, Crawford et al. 2018; see Einav and Levin 2010 and Berry and Haile 2014 for overviews). This strand of the literature focuses on overcoming the data limitations that are often present in standard economic settings, such as the absence of individual level data, unobservable product characteristics, and unobservable consumer characteristics.[3]

A second strand of the literature, primarily in developing countries, has experimentally induced price variation to estimate demand curves (Ashraf et al. 2010, Cohen and Dupas 2010, Dupas, 2014, Jessoe and Rapson, 2014, Hahn and Metcalfe, 2021). A third strand of literature has characterized demand in various settings using rich microdata on individual buyers' choices. These include, for example, Berry et al. (2003), Goolsbee and Petrin (2004), Hortaçsu and Kastl (2012),

[2] The concept of consumer surplus, or "utilite relativé," was first introduced in 1844 by French engineer Jules Dupuit. Alfred Marshall later independently reintroduced and named the concept in his 1890 publication Principles of Economics (Houghton, 1958; Svoboda, 2008).

[3] In differentiated product markets such as those studied by Berry et al. (1995) and Nevo (2000), one needs not only instruments for price, but also instruments that shift market shares through a channel other than prices (Berry and Haile 2014).

Quan and Williams (2018), and Hortaçsu et al., (2018). We also use individual-level choice data to estimate demand and welfare here. Our empirical setting provides us with a large number of regression discontinuities that gives us an estimate of the full demand curve that is less model reliant than earlier methods, even with access to large amounts of data, have had to be. To be clear, we do not view these previous approaches as being incorrect or inherently flawed. Instead, we simply see our ability to leverage the particular structure of this market, along with a massive amount of data about it, as offering a unique opportunity to estimate preferences in an important setting while relying on limited assumptions.

Our demand and welfare estimation here exploits data generated by Uber, and in particular its low-cost product UberX.[4] Three features of our data allow us to generate credible consumer surplus estimates that rely on only a limited set of identifying assumptions. The first key feature of the data is that individual consumers provide a clear signal of their interest in making a purchase (i.e. by opening the app) prior to learning the current market price. In our data, demand shocks (which interfere with demand estimation) appear to operate almost exclusively at the extensive margin of whether a consumer is in the market to make a purchase. After conditioning on a consumer's signal as to whether she is in the market for making a purchase, our further attempts to deal with endogeneity have little impact on the elasticity estimates. This approach to demand estimation could become more widely utilized by researchers and practitioners as an increasing number of marketplaces, such as airlines, hotels and large online retailers like Amazon, acquire information on who is in the market for the good.

The second key feature of our data is that Uber uses real-time pricing ("surge" pricing) to equilibrate local, short-term supply and demand. A consumer wishing to take a particular trip can face prices ranging from the base price (what we call the "no surge" or "1.0x" price) to five or more times higher, depending on local market conditions. Importantly, we observe the price offered to the consumer and whether she accepts or rejects that offer. This allows us to estimate price elasticities across a large swath of the demand curve. It is rare, outside of the auction literature, to estimate people's price elasticity of demand across the whole demand curve in a naturally occurring marketplace because historically researchers lacked such price variation. This is, however, changing as a variety of industries move to dynamic pricing.[5]

[4] The rampant growth of "peer-to-peer" transactions and the "sharing" economy have had a profound impact on many industries in recent years. A burgeoning economic literature is devoted to this topic (e.g. Cramer, 2016; Cullen and Farronato, 2014; Einav et al. 2016; Fraiberger and Sundararajan, 2015; Hall and Krueger, 2015; Farronato and Fradkin (2018).

[5] Castillo (2025) uses a structural model of Uber's Houston market to show that surge pricing raises rider surplus relative to uniform pricing, with gains across the income distribution. Our approach complements this by using surge-driven price variation to recover demand directly rather than through a structural model.

A third key feature of our data is that one of Uber's business practices – offering consumers only a discrete set of prices – creates quasi-random price variation that is useful for demand estimation. Specifically, two consumers that Uber views as facing essentially identical market conditions can receive substantially different price offers depending on whether the price they are shown is rounded up or down. Moreover, the perishability of the good means no problems with stockpiling from the change in prices (Hendel and Nevo, 2006).

Using a sample of nearly 50 million UberX consumer sessions, which represents the first 24 weeks in 2015 from Uber's four biggest U.S. markets, we estimate demand elasticities for Uber's most used service ("UberX").[6] Empirically, four basic facts emerge. First, when we attempt to naively estimate demand elasticities at the market level using aggregated data like that which would more typically available to researchers, we generate "nonsense" elasticities of demand that are greater than zero (0.238; SE=0.002). This is not unusual; if both demand and supply shocks are a source of variation, then the resulting estimates will reflect a mix of supply and demand elasticities. Second, once we condition our estimates on whether a consumer opens the app – signaling an interest in making a purchase at that point in time – our estimated demand elasticities turn negative and have similar magnitudes regardless of the sources of variation that we use in the estimation or the set of included controls. Because this comparison changes both the level of aggregation and the conditioning on app-opening at the same time, it does not by itself separate the two, but the pattern suggests that the demand shocks that clearly confound the naïve, market-level approach appear to be operating almost exclusively on the margin of whether or not a consumer is in the market. Third, estimated demand is quite inelastic. Our methodology estimates a set of price elasticities, with a median of -0.54 across the fifty estimates in Table 4; the inverse-variance-weighted average, which is dominated by the precisely estimated low-surge discontinuities, is -0.72. These magnitudes are comparable to the medium-term price elasticities estimated by Goldszmidt et al. (2020) using a field experiment where prices were randomized on the Lyft platform for eight weeks in 2016-7. Fourth, the elasticity of demand varies somewhat (but perhaps less than expected) as a function of observable characteristics such as time of day, user experience with Uber, and the presence of close substitutes (all of these elasticity results are replicated in the field experiment of Goldszmidt et al. (2020)).

These estimated Marshallian price elasticities form the basis of our consumer surplus calculations, but further assumptions are required. To compute consumer surplus, one needs to know how consumers would have responded had they faced a higher price. We do not directly

[6] The data were chosen in conjunction with Uber to be large enough to be representative, while not revealing information that may have more business sensitivity.

observe this in the data. Instead, we observe how price responsive consumers are when market conditions dictated a higher price. The set of sessions with high surge prices may, however, differ systematically in their price responsiveness from those who see low surge.[7] We deal with this complication in three ways.

First, we reweight the data to identify a subset of sessions that saw high surge prices, but whose observable characteristics (e.g., location, time of day, day of the week, and past usage of Uber) match the pool of sessions that face no surge. Second, we redo our estimates eliminating from the sample all observations where there is a positive local demand shock. The prices charged depend on the interplay of supply and demand. If the supply of drivers is low, prices can be high even though the number of requests in a given time and place are not out of the ordinary. Price spikes driven by idiosyncratically low supply are likely to provide a better counterfactual than those triggered by unusually high demand. Reweighting on observables raises the consumer surplus estimate by 23 percent, while eliminating positive demand shocks leaves the estimated demand curve essentially unchanged over the range where it can be estimated precisely.

Third, we explore whether the decision to open the Uber app is affected by predictable variations in surge. Our demand elasticities are conditional on the app being opened. If price sensitive customers do not open the app when they expect surge to be high, then only the price inelastic customers will be represented in our data at high surge levels. In practice, however, we see no evidence for such behavior. This is unsurprising given the relatively unpredictable nature of surge pricing and the fact that it takes only a few seconds to open the Uber app to check for surge.

We obtain large estimates of the consumer surplus generated by UberX. We compute the dollar value of consumer surplus from UberX rides taken in Uber's four biggest U.S. markets in 2015 (Chicago, Los Angeles, New York, and San Francisco) to be roughly $2.57 billion (SE=$141 million) annually, using our baseline assumptions. This is at least 5.6 times Uber's revenues from UberX in those cities.[8] In 2015, these cities accounted for around 42.6% of UberX US gross bookings. If we assume that consumer surplus is proportional to gross bookings, we can extrapolate to an estimate of $6.03 billion in consumer surplus from UberX in the U.S. The estimated consumer surplus is approximately 1.40 times as large as total consumer expenditures on UberX rides in these cities (about $1.84 billion). That is, for each $1 spent on UberX, we

[7] Note that the same consumer opening the app under different circumstances may have different likelihoods of making a purchase and different sensitivities to price. Thus, we focus on sessions as our unit of analysis, not individuals.
[8] This assumes a 25% commission of gross fares reserved for Uber. This percentage likely overstates Uber's actual average commission rate on the trips represented.

estimate consumers receive $1.40 in surplus. Because about 84 percent of this surplus accrues to riders who paid the base price, the estimate depends chiefly on whether riders observed near surge thresholds are representative of base-price riders. Reweighting to match base-price riders on observables raises the estimate to about $3.16 billion; Table 7 collects the estimates under all of the alternative assumptions we consider, which range from roughly $2.4 billion to $3.4 billion. These estimates of consumer surplus are large relative to the likely gains or losses experienced by taxi drivers as a consequence of Uber's entrance into the market (Cramer 2016). This increase in consumer surplus of $1.40 for each dollar spent is larger than what has been found previously for automobiles (Berry et al., 1995; Petrin, 2002).

From a public policy perspective, our consumer surplus estimates have two shortcomings. First, they are derived from short-run demand elasticities, but any policy decision is likely to be interested in long-run consequences.[9] However, our short-term price elasticities of demand are consistent with those medium-term elasticities, the latter estimated from natural field experiments with randomized prices changes that are consistent over an eight-week period in the same cities as ours (Goldszmidt et al., 2020).[10] Second, our estimates miss the consumer surplus associated with other ride-sharing products (both those offered by Uber and by other ride-sharing companies), as well as consumer benefit or harm resulting from responses of the taxi cab industry to Uber's entry. We discuss in the concluding section what economic theory tells us about mapping from the numbers we are able to credibly estimate to the numbers that are of greatest economic interest. In principle, our estimates can be used as an input to a host of important public policy questions, such as optimal Pigouvian congestion prices and the impact of regulating ride-sharing services. Throughout, our estimate should be read as the surplus consumers would lose if UberX alone were unavailable for a short period, with all other transportation options, including other Uber products and competing ride-share services, held fixed.

Although very different methodologically from our paper, Buchholz (2016) is the most similar prior work in terms of goals. Buchholz (2016) estimates a dynamic spatial equilibrium model of New York City taxi cabs to assess the efficiency cost of existing regulations. He concludes that

[9] The price variation we exploit is highly transient, and the set of competitors is fixed. Thus, the appropriate interpretation of our estimate is roughly, if Uber's system malfunctioned and Uber were therefore unavailable for a day, how much would consumers suffer? (The answer would be 1/365th of our annual consumer surplus number: about $7.0 million per day in our four cities, or about $16.5 million nationally on our extrapolation.)

[10] In a randomized experiment offering Uber riders in Cairo discounts of up to 50 percent for three months, Christensen and Osman (2023) find a much larger usage response, consistent with more elastic demand for ride-hailing where it substitutes for lower-cost transport options and incomes are lower.

efficient two-part tariff pricing and a directed matching technology would deliver welfare gains of over $2 billion annually in New York.[11]

The remainder of the paper is structured as follows. Section 2 provides background on Uber. Section 3 presents a simple model that motivates our empirical specifications and states the assumption needed to translate our estimates into consumer surplus. Section 4 describes the data, presents our regression-discontinuity estimates of demand elasticities, and discusses the validity of the design and the limits of our inference. Section 5 translates the elasticities into consumer surplus and examines how the estimate changes under alternative assumptions, which Section 5.8 summarizes. Section 6 concludes with a discussion of how our estimate maps into policy-relevant counterfactuals. The appendix contains supplementary specifications.

2. Background on Uber

Uber is a technology company founded in 2009, which created a smart phone application that matches and handles payments between consumers seeking rides and Uber's "driver-partners." Uber's service has proven extremely popular, growing dramatically in terms of both geography and volume.[12]

To use Uber, a consumer downloads the app onto her smartphone for free. When seeking a ride, the consumer opens the Uber app and sees something akin to the screenshot in Figure 1. There is a map of the local area, a display of driver-partners in the area available to provide rides, and an estimate of how many minutes it will take the nearest vehicle to reach the consumer's location. Uber offers a number of different products, as shown near the bottom of the screen in Figure 1. The user is able to scroll between those products. If a consumer places an order, driver-partners are sequentially given the opportunity to accept that order until one does so. That driver-partner then picks up the rider and drops her off at her desired location. Uber defines a user product-session in which the user opens the app, culminating either in the user ordering a ride or electing not to order a ride.[13] Throughout this interaction, Uber records all actions taken on the app as well

[11] More specific to the literature on taxis, there are a very rich set of papers that have attempted to understand the supply side of the taxi market (see Camerer et al., 1997; Farber, 2005, 2008, 2015; Crawford and Meng, 2011).

[12] City governments have had varying reactions to Uber's operations, which dramatically alter the status-quo of transportation. Historically, for-hire transportation has been heavily regulated, usually via a taxi medallion system (see Frankena and Pautler, 1984). The incumbent taxi cab providers, not surprisingly, have been hostile to Uber's presence.

[13] A session is Uber's best attempt to identify a consumer's decision as to whether or not to make a purchase. For simplicity with our focus on UberX, we use "session" to refer to a product-session of UberX interaction unless otherwise specified. Technically, if the rider interacts with another Uber product at a similar time, this creates a separate product-session. A session is defined as a period of, not necessarily continuous, use between opening the app to UberX and either requesting a ride or ceasing to use the app for a period of 30mins. Thus, multiple closings and

as certain background information relevant to the transaction. These data are collected and stored regardless of whether or not the session ends with a purchase.

Uber offers several products, which differ in terms of the types and size of cars, whether the ride is shared with other passengers, and the price. Our focus is on UberX, the core product that represented almost 80 percent of all Uber rides during the time period of our sample. With UberX, a rider summons a driver-partner who drives her own private vehicle and delivers the rider to the desired location without stops.[14]

Uber's base pricing system has components similar to standard cab pricing systems in which each city and product has a fare defined by price per mile, price per minute, a fixed fee, and a minimum total fare.[15] In contrast to regulated cabs, Uber also utilizes a dynamic pricing system, called surge pricing, on many of its products.[16] Uber's surge algorithm monitors rider demand and available driver supply and institutes a multiplier on the base price when demand outstrips supply at the base price.[17] This pricing system helps increase supply at times of high demand, and allocate rides to riders who value them most highly (Hall et al. 2015; Castillo 2025), but does not itself clear the market. The time a customer waits for a ride (and conversely the fraction of time a driver-partner has a rider in the vehicle) also varies with supply and demand.

Around 21% of UberX sessions in our dataset have some surge price exceeding 1.0x. Figure 2 presents the observed distribution of UberX sessions in our data for surge prices greater than 1.0x. The number of sessions between 1.2x (the lowest surge) and 1.5x are roughly equal. 1.5x is somewhat overrepresented due to less than perfect accuracy in excluding sessions where the surge price was altered by Uber business rules. Beyond that point, the number of sessions decreases monotonically with the level of surge. Roughly 4.1 percent of surge sessions involve surge levels at or above 3.0x; 0.65 percent involve a surge greater than or equal to 4.0x.

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openings of the app in a short period of time do not generate many sessions. The median elapsed time of a session is 31 seconds.

[14] UberXL is identical to UberX, except that the requested vehicle must accommodate at least six passengers. UberBlack and UberSUV work just like UberX and UberXL respectively, except that the driver-partners are Transportation Charter Permit (TCP) licensed and the vehicles driven meet a higher set of standards. UberPool differs from the products above in that multiple riders are picked up en route by the same driver-partner, increasing the expected travel time and changing the user experience. UberTaxi differs from the other products in that it only summons licensed taxi drivers in licensed taxis, and the fares match the regulated taxi fares in the city.

[15] During the period of this study, each rider is also charged a $1 Safe Ride Fee, since renamed to Booking Fee, used specifically for ensuring the safety of riders and driver-partners on the Uber system. The Safe Ride Fee is not included in the base fare (and hence is unaffected by surge). An example fare can be found in the Appendix.

[16] UberPool is priced according to a different system which is beyond the scope of this paper.

[17] Surge prices are always greater than or equal to 1.0, i.e., the price is never lowered below the base fare, even when market conditions suggest it should.

Although consumers are only shown a limited number of discretized surge levels, the algorithm generates a continuous measure of surge (which we call the "surge generator" or "generator"). This proves to be extremely useful for our identification strategy. Two customers who have nearly identical surge generators (i.e., face nearly identical market conditions), but who happen to be on opposite sides of a pre-defined cut-off, face discretely different surge prices.

Figure 3 shows the frequency of surge by hour of the day and day of week for UberX, displayed as a heat map. The darker the color, the greater the share of trips involving surge pricing. Surge pricing is most likely to occur late at night (especially on weekends) and during the morning rush hour, but even at those times surge is invoked in less than half of the sessions. Surge pricing is less tightly linked to intuitions about demand than might be naively expected. This is because driver-partners seeking more revenue adjust their labor supply in response to predictable demand shocks.[18]

3. The Model

We develop a model that motivates the logic behind our empirical specifications. Suppose a potential UberX rider's utility level from purchasing a ride in a particular location at a particular time is given by

$$u_{ij} = \delta_j - \alpha_i m_j p_i - \beta_i w_j + \epsilon_{ij} \qquad (1)$$

where i indexes riders, j indexes time-location settings, δ_j is the mean utility of an UberX ride at a specific location-time, α_i is rider i's price sensitivity, p_i is the baseline fare rider i faces for their desired ride, m_j the surge factor (which is common across all riders at a given location-time), w_j the waiting time for a ride to arrive (also common to a given location-time), β_i is rider i's sensitivity to waiting time, and ϵ_{ij} is a mean-zero random utility term.

Dividing through by rider i's base fare in equation (1) yields:

[18] High surge prevalence in the early morning is consistent with compensating differentials for driver-partners who have higher disutility from working those hours (or perhaps associated with the clientele seeking late night rides). Also consistent with this hypothesis is that driver-partners earn more on average on weekends than weekdays (both because of higher surge and more rides per hour.) There are some dimensions across which drivers cannot easily substitute, for example, across cities. Substantial differences in surge persist along those dimensions. For example, 14.4 percent of all New York City sessions involve surge, compared to 24.7 percent of Chicago sessions.

$$\tilde{u}_{ij} = \frac{\delta_j}{p_i} - \alpha_i m_j - \frac{\beta_i}{p_i} w_j + \frac{\epsilon_{ij}}{p_i} = \tilde{\delta}_i - \alpha_i m_j - \tilde{\beta}_i w_j + \tilde{\epsilon}_{ij} \qquad (2)$$

where $\tilde{\delta}_i \equiv \frac{\delta_j}{p_i}$, $\tilde{\beta}_i \equiv \frac{\beta_i}{p_i}$, and $\tilde{\epsilon}_{ij} \equiv \frac{\epsilon_{ij}}{p_i}$ can be interpreted as values of rider utility per dollar of baseline fare.

Equation (2) can be rewritten as

$$\bar{\delta}_j - \bar{\alpha}_j m_j - \bar{\beta}_j \tilde{w}_j + \left(\tilde{\delta}_i - \bar{\delta}_j\right) - \left(\alpha_i - \bar{\alpha}_j\right) m_j - \left(\tilde{\beta}_i - \bar{\beta}_j\right) w_j + \tilde{\epsilon}_{ij} \qquad (3)$$

where $\bar{\delta}_j$, $\bar{\alpha}_j$, and $\bar{\beta}_j$ have been defined respectively as the average levels of $\tilde{\delta}_i$, α_i, and $\tilde{\beta}_i$ for a given location-time. They are indexed by *j* accordingly. We define the mean-zero error term $\varepsilon_{ij} \equiv \left(\tilde{\delta}_i - \bar{\delta}_j\right) - \left(\alpha_i - \bar{\alpha}_j\right) m_j - \left(\tilde{\beta}_i - \bar{\beta}_j\right) w_j + \tilde{\epsilon}_{ij}$ and, as an approximation, treat it as having the standard T1EV form. Thus the utility per dollar spent for riders at a given location-time *j* is:

$$\tilde{u}_{ij} = \bar{\delta}_j - \bar{\alpha}_j m_j - \bar{\beta}_j w_j + \varepsilon_{ij} \qquad (4)$$

Riders choose between UberX and an outside good that is assumed to offer a mean utility normalized to zero, plus a T1EV term:

$$u_{i0} = \varepsilon_{i0} \qquad (5)$$

These assumptions yield the standard logit choice model; namely, the probability that rider *i* chooses UberX at location-time *j* is

$$\Pr\{i \; rides \; UberX \; on \; occasion \; j\} = \frac{\exp\left(\bar{\delta}_j - \bar{\alpha}_j m_j - \bar{\beta}_j w_j\right)}{1 + \exp\left(\bar{\delta}_j - \bar{\alpha}_j m_j - \bar{\beta}_j w_j\right)}$$

The logit form is intended only to motivate our empirical specifications; we do not estimate it. Our elasticities come from linear-probability regression discontinuities at each surge threshold, and our consumer surplus calculation in Section 5 chains these local elasticities together without imposing a functional form on demand. Two points about the model should be kept in mind. First, because ε_{ij} contains the term $(\alpha_i - \bar{\alpha}_j)m_j$, its dispersion rises with the surge multiplier, so the T1EV assumption can hold only approximately; nothing in our estimation relies on it. Second, what our approach does require concerns extrapolation. The elasticities we measure come from sessions whose surge generator lies near each threshold, but the consumer surplus calculation applies them to all purchasing sessions, including those quoted 1.0x, which account for about 84 percent of our

estimated surplus (Table A2). The key identifying assumption is therefore that the willingness to pay of sessions near each threshold is the same as that of the sessions whose surplus we compute, either unconditionally or conditional on observables. Section 3.1 explains how the discontinuities identify the local elasticities, and Section 5 examines this assumption directly.

3.1. Identification and Specification Testing through the Lens of Our Model

We now discuss our identification strategies, using the model to motivate the logic behind our empirical specifications. To identify the price sensitivity of demand, one needs shifts in prices that are orthogonal to changes in demand other than through price. Our core specification leverages the discreteness of the surge factor as a source of identification. This discreteness results from Uber's matching algorithm in a way we describe here.

In absence of surge pricing, the market for Uber rides would clear via wait time w_j. If rider demand considerably exceeded the supply of available drivers at the base fare level, wait times would increase until the queue of excess demand clears, either through eventual accommodation of demand or through potential riders choosing the outside option. To avoid long wait times and the concomitant loss of revenue and goodwill, the Uber platform uses information gathered from potential riders and drivers in the vicinity of j to set m_j to try to equalize the quantity of rides that riders demand and drivers supply. This equalization does not happen exactly due both to errors in Uber's modeling of market conditions and their decision to set the surge factor at discrete intervals and below certain maximum levels. As a result, residual mismatches in quantities clear through wait times, though much smaller ones than would be the case at the base fare.[19]

Before explaining how Uber's treatment of the surge factor lets us identify demand in our data, we first note an endogeneity typically endemic in demand estimation that we do *not* face. Our demand specification estimates probability of purchase conditional on a potential rider being in the market for transportation at occasion j. It takes as given the total number of potential riders at j. In many demand estimation settings, this "size of the potential market" is part (often a large part) of the variation in realized quantity demanded. As such, estimating how it shapes demand would require appropriate instruments. Our estimation, however, conditions out this variation, needing only to estimate the conditional probability of purchase. For example, if the potential demand for rides outside a stadium is 5,000 on nights when there is a game and 50 when there is

[19] Note that because riders observe discretized expected wait time when they choose to request a ride, their purchase decision is conditional on that value. In other words, the demand curve we observe from riders' behavior is conditional on expected wait time. We can accomplish this conditioning statistically because we can obtain estimates for $\bar{\beta}_j$, riders' average disutility of wait time, using our separate wait time dataset (instrumenting for the discrete measure of wait time riders see with Uber's continuous underlying measure).

not, modeling the total quantity of rides demanded would require instruments that reflect shifts in the total potential supply of rides. Our approach, however, conditions on overall potential market size. It asks, effectively, what the probability is that one of those 5,000 (or 50, as the case may be) potential riders in fact chooses Uber. That probability could well vary systematically across situations like whether there was a game or not that night at the stadium. This is embodied in δ_j, and to estimate demand consistently we need price variation that is orthogonal to it (we argue below why we believe this is the case). However, our conditional approach does not need supply shifts that are orthogonal to the variation in the overall number of potential riders.

Now let us explore how Uber's treatment of the surge factor identifies (conditional) demand in our setting. Suppose that, given the state of demand and supply in location-time *j*, that the surge factor that would exactly equilibrate demand and supply at target wait time w_j^T is $\widehat{m}_j$. However, due to modeling error and the restrictions Uber puts on its possible values, the actual surge factor m_j will either be somewhat below $\widehat{m}_j$, so $m_j = \widehat{m}_j - \Delta_l$, or somewhat above $\widehat{m}_j$, so $m_j = \widehat{m}_j + \Delta_h$, where $\Delta = \Delta_l + \Delta_h$ is (for example) the step size of the surge factor. Clearly in equilibrium, $\widehat{m}_j$ will generally be a function of shifts in demand and as such not a source of identification of riders' price sensitivities. However, if the realizations of Δ_l and Δ_h are orthogonal to demand shifts—and given Uber's surge pricing equilibrium, we believe there is a strong argument they are—then differences in quantity demanded driven exclusively by the discrete variation in the surge factor Δ will be due only to price effects and not any other demand shifts. This is the logic of our regression-discontinuity-style identification; Δ in effect serves as an exogenous instrument for price variation in the neighborhood of the discrete price change.

The three approaches we employ to deal with possible changes in the composition of demand across the surge distribution can also be interpreted using this framework. Under our assumption that the demand parameters $\bar{\delta}_j$, $\bar{\beta}_j$, and $\bar{\alpha}_j$ are uncorrelated with the equilibrium surge factor $\widehat{m}_j$, our estimates fully characterize rider demand. The three approaches are different ways to check indirectly if the assumption holds in the data. The concern is that the realized values of $\bar{\delta}_j$ and $\bar{\alpha}_j$ might vary systematically with the equilibrium surge factor $\widehat{m}_j$.

Our matching strategy presumes that demand parameters are functions of observable variables X_j about the location-time occasion, so for example $\bar{\delta}_j = \bar{\delta}_j(X_j)$ and $\bar{\alpha}_j = \bar{\alpha}_j(X_j)$. We can flexibly characterize these functions to see if more closely matching observations at different surge levels on observables greatly affects our estimates. Suppose for example that a certain (observable) type of potential rider is systematically less likely to encounter surge rates above 1.0 but has a relatively high probability of purchase (a high $\bar{\delta}_j$). When estimating demand using

variation around higher surge rates, we would be systematically undermeasuring willingness to pay because these high-$\bar{\delta}_j$ riders would be less populous in such situations than on average. Our reweighting approach adjusts for this by giving greater weight to the riders of the high-$\bar{\delta}_j$ type that we *do* observe at higher surge levels.

Our bucket-reweighting method starts by essentially estimating the joint distribution of observables and equilibrium surge factors, $g(X_j, \hat{m}_j)$. In practice, this means partitioning the sample by categories of X_j and then computing the fraction of observations in each category corresponding to every observed surge factor m_j. Next, to obtain a counterfactual estimate of what estimated demand would have looked like if sessions that in reality saw (for example) $m_j = 1.5$ instead had observables mirroring those in cases where $m_j = 1.0$, we estimate price sensitivities using the discontinuity around $m_j = 1.5$ by up-weighting observations that are in parts of the joint distribution with high frequency of $m_j = 1.0$ relative to $m_j = 1.5$. We are thereby constructing a weighted distribution of demand responses around $m_j = 1.5$ that "looks" less like the marginal distribution $g(X_j|m_j = 1.5)$ and instead more like $g(X_j|m_j = 1.0)$.

Our second strategy for addressing rider composition issues uses variations in surge rates driven by supply conditions. The logic is straightforward. While our discontinuity approach should deliver exogenous local price variation to identify riders' price sensitivities around any given surge factor, to the extent that the level of the surge factor is determined by demand shifts, then there can be systematic differences in the composition of riders and their preferences *across* observed surge factors. So if for example in situations when there are a large number of riders considering ordering rides, and those riders have particularly high values of utility conditional on ordering a ride, using only supply-side-driven surge values addresses that potential problem. (This follows the above logic that demand parameters can be a function of observables; here, the observables would be rider characteristics.)

By using surge rate variation that arises from supply differences—for example, shifts in drivers' nonpecuniary returns to driving on a particular occasion, $\tilde{\mu}_j$, like variations in their outside options—we are essentially holding $\bar{\delta}_j$ and $\bar{\alpha}_j$ (and therefore X_j) fixed as we compare riders' price responsiveness across different surge rates. (This presumes that the supply shifts $\tilde{\mu}_j$ are uncorrelated with $\bar{\delta}_j$ and $\bar{\alpha}_j$, which we find plausible.) In this way, the logic of this strategy is very much like the standard logic for estimating demand with supply-side instruments. It is just that here, rather than using the supply-side instruments to obtain price sensitivity estimates, we are using it to control for composition changes in demand parameters that might exist across markets with different equilibrium prices.

Our third strategy tests whether our elasticity estimates systematically differ across periods of predictably and unpredictably high surge to test whether potential riders who are on average more price sensitive are less likely to open the app (and therefore less likely to ride) on occasions where surge is expected. This test is based on the idea that, if such self-selection occurs, this would be reflected in demand parameters because $\bar{\delta}_j(X_j|expected\ surge) \neq \bar{\delta}_j(X_j|unexpected\ surge)$ and $\bar{\alpha}_j(X_j|expected\ surge) \neq \bar{\alpha}_j(X_j|unexpected\ surge)$. The concern is that, effectively, opening the app predicts systematic differences in demand that are correlated with realized price variation. We find only a small and statistically insignificant difference, which is directionally consistent with modest selection into opening the app. Section 5.4 gauges what selection of this size would imply for consumer surplus.

4. Data, Identifying price elasticities, and estimation

We focus our analysis on UberX because it is the only Uber product that has both substantial scale and frequent surge pricing, which is central to our identification strategy. Our primary data set covers all UberX sessions in Uber's four largest U.S. markets (San Francisco, New York City, Chicago, and Los Angeles) over the period January 1, 2015 to June 17, 2015.

Our unit of analysis is a customer session, a company-defined measure that captures a particular consumer trying to order a particular ride. There are approximately 54 million UberX sessions in our raw data. For each session, we observe the surge price, underlying surge generator (which the consumer does not see), Uber-defined geographic region, time and date, anonymized rider id, Uber's prediction of expected wait time (consumers see wait time in minutes), product, and the ultimate decision of the rider whether to request a car.[20] Uber has a variety of business rules in place that override the surge algorithm in certain cases, primarily to prevent riders and drivers from experiencing very sharp price changes. In our main analysis, we discard these rider sessions, which make up roughly 11.5% percent of the sample, leaving approximately 48 million observations in our entire sample.[21] Summary statistics for this sample are presented in Table 1. We present statistics for the entire sample (column 1), as well as for three mutually exclusive and exhaustive subsets of the data: rider sessions with baseline pricing, (i.e. surge equal to one, in column 2), moderate surge (surge between one and two, in column 3) and high surge (column 4).

[20] Our methodology treats a car request as a purchase. Although there are times where either the driver or the rider decides to subsequently to cancel the car, we do not consider these cases as meaningfully informing our interpretation of the rider's decision to purchase.

[21] We will, however, use those excluded observations for supplementary analysis.

A number of patterns emerge in Table 1. The entries in the top row highlight the wide range of prices that consumers are exposed to due to surge. Relative to baseline prices in Column 2, the same trip costs a consumer roughly 50 percent more on average with moderate surge (Column 3) and is two and one half times more expensive on average when high surge is in effect (Column 4). Expected wait times are not highly correlated with surge. Although surge kicks in when demand is high relative to supply (implying long expected waits), high surge reduces demand and increases supply, equilibrating wait times. Purchase rates, shown in the third row, decline from 62 percent in Column 2 to 39 percent in Column 4 as surge rises. In percentage terms, however, purchase rates decline less than prices rise. This pattern foreshadows the consistently inelastic demand estimates found throughout the paper.

The remaining rows of Table 1 describe the distribution of our observations across city, time of the day and week, by the number of Uber trips made by the consumer. Our sample is split relatively equally across the four cities, with high-surge trips relatively less frequent in New York and more frequent in Chicago. As shown in Figure 3 above, surge most frequently occurs during rush hour and during weekend "party" hours. Most of the observations in our sample are associated with frequent Uber riders. The bulk of our data comes from frequent users of the product (more than eight rides in the period), and these frequent users are overrepresented during high surge periods.

4.1 From Naïve to Conditional Demand Estimates

Before turning to our actual identification strategy, which is based on plausibly exogenous price shocks at the individual level, for purposes of comparison we present naïve, market-level demand estimates from regressions that take the form:

$$ln\ Requests = \beta \ln(Price_{mt}) + \gamma ETA_{mt} + \gamma_{city} + \lambda_{timeofday} + \varepsilon$$

where *m* captures the 783 geographic market windows in which Uber offers a uniform price and *t* corresponds to a particular five-minute window in which a price is offered. At this time, Uber had 13,959,454 market × time windows across the four cities in our sample. *The dependent variable, log requests, is the log of* the number of rides requested in a particular market window in a specific five-minute window; *Price* is the level of surge that all consumers in a given market window and time period faced. *ETA* is the average estimated wait time reported to consumers. Also included in the most fully specified models are city fixed effects and seven indicator variables that partition the week to capture systematic variation in surge levels.

The results from these aggregated regressions are presented in Table 2. The top row shows the coefficient on the variable of interest -- logged price -- along with the corresponding standard error. The remaining rows report coefficients on the additional covariates. Each column corresponds to a different specification, with the number of controls increasing moving from left to right. Column 1 includes no controls. Column 2 adds expected wait time. Column 3 includes city fixed effects. Column 4 adds time of day/day of week indicators. There are nearly 14 million observations per column. In all cases, the regressions produce tightly estimated positive elasticity estimates ranging from 0.343 to 0.238. These positive estimates, which are inconsistent with a downward sloping demand curve, appear to capture a mix of shocks to supply and demand. This, of course, is not surprising, and motivates the identification strategy we adopt below.

These estimates differ from the individual-level estimates that follow in two respects at once: they aggregate to the market level, and they do not condition on consumers having signaled interest by opening the app. The contrast between Table 2 and Table 4 therefore shows that the combination of the two changes reverses the sign, but it does not by itself establish how much of the reversal is due to conditioning on app-opening rather than to the use of individual-level, discontinuity-based variation. We return to this point in the conclusion.

4.2 Identifying Price Elasticities

If the degree of surge pricing faced by a consumer on a given trip were generated at random, then all that would be required to trace out a demand curve would be to compute the share of Uber searches culminating in a ride at each level of surge pricing. With randomization, if 70 percent of searches lead to a transaction at the base price, but only 63 percent of searches lead to trips when the price is ten percent higher (1.1x surge), then we could assume that people who received surge 1.0x would also have requested trips at a rate of 63 percent, had they been quoted the 1.1x price. This would imply that the elasticity of demand would be one on this part of the demand curve (i.e. a 10 percent reduction in the share of people who accept the offer--from a 70% purchase rate to a 63% purchase rate--is associated with a 10 percent increase in price).[22] Similar comparisons of ride completion rates at higher prices would trace out demand over whatever range of prices consumers faced. Combining these elasticity estimates with the actual quantity purchased at 1.0x

[22] In this example, the 63 percent of consumers who demonstrated a willingness to pay of 1.1x, reveal that had they only been asked to pay the base price, they would have received a consumer surplus of at least 10 percent of that base price. The 7 percent of customers who refuse to transact at 1.1x, by this same logic, reveal a consumer surplus when transacting at the base price that is less than 10 percent of the base price.

surge yields the demand curve for customers offered 1.0x surge, as well as an associated consumer surplus.

In practice, the surge price that consumers face is not random; it reflects local demand and supply conditions. There is, however, a component of Uber pricing that is random from a consumer's perspective. Uber calculates each surge price to an arbitrary number of decimal places, but consumers are presented with discrete price increments (e.g., the lowest surge price is 1.2x, or 20 percent higher than the base price) to facilitate a simple, easy user experience.[23] Market conditions are nearly identical when the algorithm suggests a surge of 1.249x and when it suggests a surge of 1.251x, but in one case consumers face a 1.2x surge and in the other case they face a 1.3x surge. This provides the opportunity for regression discontinuity (RD) analysis, which allows us to estimate local elasticities of demand across the full range of surge prices.[24] A complicating factor in our analysis is that the expected wait time a consumer faces systematically changes at the price discontinuity; at the higher price, fewer consumers complete the purchase, so there are more vehicles available to service those customers who do buy. Fortunately, we observe in our data the same information that is shown to the potential customer regarding wait times (i.e. the expected wait time between making a request and the driver arriving). Thus, we can control directly for wait time in our analysis. Additionally, the expected wait time algorithm used by Uber is continuous, but is rounded to whole minutes when presented to customers. This allows us to use an RD design for identifying the causal impact of expected wait time on purchases and thus to even more convincingly purge any impact of wait time differences from our price elasticity estimates.

Figures 4 provides a visual example of our identification strategy, showing how purchase rates vary as a function of the surge generator over the range 1.15 to 1.35. The horizontal axis is the surge generator. The vertical axis is the percent of all sessions that convert into purchases within each bucket. Each point in the graph represents the aggregation of all observations falling into a particular generator bucket, with each bucket having a width of 0.0025. The vertical line at 1.250 represents the point at which price jumps discontinuously from 1.2x to 1.3x. To the left of that vertical line, all customers face a price of 1.2x; to the right of the line price is 1.3x. Moving from left to right towards the vertical line in the figure, purchase rates rise slightly, before dropping sharply at the price discontinuity. To the right of the vertical line, purchase rates continue a slight

[23] For example, Uber might estimate that the appropriate multiplier is 1.61809, but for easy interpretation, they would charge the customer 1.6x.

[24] Additionally, there are Uber business rules that sometimes cause prices to be far below what the surge algorithm recommends, allowing us also to analyze consumer behavior when the differences between the surge level and the surge generator are larger.

positive trend.[25] The purchase rate falls approximately 3 percent at the discontinuity; price rises by 8.3 percent (the percent difference between 1.2x and 1.3x), for an implied price elasticity of roughly -0.36 (i.e., -3/8.3).

Figure 5 displays the relationship between purchase rates and price discontinuities across a broader range of surge values. Once again, the surge generator is on the horizontal axis and the purchase rate is on the vertical axis. Each bar in the figure aggregates all sessions within a .01 surge generator window. Red bars lie just to the left of price discontinuities; yellow bars are just to the right of price discontinuities. All intermediate price ranges are shown in grey. In each of the 13 cases, the red bars are higher than the adjacent yellow bars, indicating that purchase rates fall as price discontinuously jumps. In contrast, at any given price level (the gray bars between a yellow and a red bar), purchase rates exhibit no obvious pattern of increase or decline.

To estimate a demand curve, one needs to hold everything else constant, other than price. To test whether this is indeed the case at our surge-driven price discontinuities, we estimate equations of the form:

$$\begin{aligned} Outcome = \ & \alpha + \theta * Window * Post + \beta_2 * Window + \beta_3 * (1 - Window) * Post \\ & + \beta_4 * Wait + \beta_5 * (1 - Post) * Generator + \beta_6 * Post \\ & * Generator + \varepsilon \end{aligned} \qquad (2)$$

where *Outcome* represents an outcome of interest (e.g., was a purchase made, was the session during rush hour, etc.). *Window* is an indicator as to whether the observation lies close to a price discontinuity, *Post* is an indicator taking the value of one when the observation is to the right of the price discontinuity, *Wait* is the expected wait time, *Generator* is the continuous measure of surge produced by the Uber algorithm.[26]

Table 3 reports the results of these regressions for a wide range of outcomes. Each row represents a separate regression on a different dependent variable. We report only the key coefficient θ, which captures the average difference in the outcome variable just after a discontinuity versus just before (using a window on either side of the discontinuity of 0.01), controlling for other factors. The first row corresponds to the request rate (i.e., the share of observations in which a purchase is made). As would be expected given the results in Figures 4 and 5, request rates exhibit a clear decline at the price threshold where the discontinuity occurs: (a coefficient of -0.0201 with a t-statistic of 20). Note, however, that expected wait times also drop

[25] The slight positive trend moving from left to right is not a consistent pattern over the range of surge generator levels.
[26] When our outcome is expected wait time, we exclude the control for wait time in the regression.

discontinuously at the surge thresholds: customers wait an average of 0.129 minutes (i.e., approximately 8 seconds) less after the threshold. This result is not surprising; it occurs as a consequence of purchases declining at the threshold as well as the increased incentive for drivers to make pick-ups in this area because of the higher prices. Failure to take these differences in wait times into account will bias our price elasticities towards zero, since higher prices are correlated with lower wait times.

Empirically, we deal with this complication in two ways. First, we control for expected wait times in some of our specifications. Second, in other specifications we not only control for expected wait times, but also instrument for expected wait times to deal with concerns of endogeneity. Conveniently, Uber's measure of expected wait time is measured in seconds, but consumers are only shown wait times rounded to whole minutes. Thus, we can exploit the sharp discontinuities in wait times displayed to consumers to identify their sensitivity to waiting, just like we use the price discontinuities to identify sensitivity to price.[27] Importantly, the wait-time discontinuities are essentially uncorrelated with the price discontinuities we exploit. The t-statistic on our wait-time instrument in the first stage regression is around 20 (regression and detailed description in Appendix: First-stage expected wait time regression).

In contrast to wait times, the other observable characteristics vary only weakly at the price thresholds, as would be expected.

Validity of the design. Three features of the setting support the discontinuity design. First, consumers never see the surge generator, only the rounded multiplier, so they cannot sort themselves to one side of a threshold. The thresholds themselves are fixed display increments rather than values chosen in response to local conditions. Second, Figure 5 provides a form of placebo evidence: within each price level, where the generator varies but price does not, purchase rates show no systematic pattern, while every one of the 13 price thresholds shows a drop. Section 5.6 confirms formally that, holding price fixed, purchase rates move only slightly with the generator. Third, Table 3 shows that observable characteristics other than wait time vary only weakly across the thresholds. Two limitations should be noted. All estimates use a single window of 0.01 on each side of each threshold, so we cannot show how sensitive they are to the choice of bandwidth. We also do not report a formal test for discontinuities in the density of the generator at the thresholds.

[27] Uber primarily stores wait times as minutes. Accessing wait times in seconds turns out to be time consuming and difficult. With substantial effort, we were able to pull two weeks of this data. As it turns out, the estimated impact of longer wait times on purchase rates is virtually identical when estimated using OLS or when we instrument for wait time using discontinuities, so instrumenting has no discernible impact on either the coefficient on wait time in a price regression or on our price elasticity estimates.

Inference. The surge generator, and therefore the price, is common to all sessions in a pricing zone during a given five-minute window, and sessions within a zone-window share unobserved shocks. The roughly 48 million sessions in our sample therefore contain fewer independent draws of price than the session count suggests. Our reported standard errors treat sessions as independent and are likely to understate the true sampling uncertainty; the same applies to the bootstrap standard error for consumer surplus in Section 5. Because even the number of zone-by-window cells is very large (nearly 14 million, as reported in Table 2), we do not expect this to overturn the main findings, which rest on large and consistently signed discontinuities, but the reported standard errors should be read as lower bounds.

4.3 Estimating Price Elasticities

In principle, we can estimate price elasticities simply using the observed discontinuities in purchase behavior around the price jumps. We also estimate specifications that control for a range of observable characteristics that might influence demand. We run a separate regression for each price discontinuity, which is identical to equation (1) above with purchase as the outcome, but with the addition of fixed effects for city and our eight time of the week indicators:

$$\begin{aligned} Purchase \;=\; & \alpha + \theta * Window * Post + \beta_2 * Window + \beta_3 \\ & * (1 - Window) * Post + \beta_4 * Wait + \beta_5 * (1 - Post) \\ & * Generator + \beta_6 * Post * Generator + FE(city) \\ & + FE(hour\ and\ day) + \varepsilon \end{aligned} \tag{3}$$

where *Purchase* is an indicator variable for whether or not a session ends in a purchase, *Window* is an indicator as to whether the observation lies within 0.01 of a price discontinuity, *Post* is an indicator taking the value of one when the observation is to the right of the price discontinuity, *Wait* is the expected wait time, *Generator* is the continuous measure of surge produced by the Uber algorithm. Fixed effects for city and our eight "*Hour and day"* categories are also included. For each price discontinuity, we include all sessions that are quoted the price falling on either side of the discontinuity (e.g., for the price discontinuity between 1.5x and 1.6x surge pricing, all observations at 1.5x and 1.6x are included). The price elasticities are identified only using the observations within the window around the discontinuity; the remaining observations are nonetheless useful for estimating the other coefficients.[28]

[28] Due to the large number of fixed effects being used, we chose the linear probability models over using logits or probits, although we recognize the potential shortcomings of such an approach.

The coefficient of interest is θ which captures the drop in purchase rates at the price discontinuity, controlling for other factors and allowing for a different linear trend in the generator on the two sides of the threshold. To transform these values into elasticities, we use the definition of a price elasticity:

$$Elasticity = (\%\Delta\ in\ Quantity/\%\Delta\ in\ Price) = ((\theta/Purchase\ Rate)/ \%\Delta\ in\ Price) \quad (4)$$

where *Purchase Rate* is the share of sessions that result in purchases at a given price. Since the purchase rate and percent change in price are directly observed in the data, there is a direct mapping between θ and the price elasticity.

Table 4 presents these elasticity estimates. Each row in the table corresponds to a different surge-driven price discontinuity.[29] For instance, the first row reflects the discontinuity where prices jump from 1.0x to 1.2x; the second row is the jump from 1.2x to 1.3x. At higher surge levels, we have fewer observations and consequently our estimates become less precise. Therefore, we report pooled estimates for all discontinuities from 1.9 to 2.3, 2.4 to 3.0 and 3.1 to 5.0[30] At these higher surge levels, we generate separate estimates around each discontinuity, and then we present an inverse variance-weighted average of the estimates. Each column in Table 4 corresponds to a different specification. Column 1 shows the results from our RD approach, absent any other controls. Columns 2 adds expected wait time as a control. Column 3 instruments for expected wait time.[31] Column 4 adds controls for time of week and city as controls to the specification in Column 3.[32]

Four notable patterns emerge in Table 4. The first is how strikingly different the estimates are from the naïve market-level estimates in Table 2. This suggests that the identifying variation at the market level is heavily driven by demand shifts. The reversal of sign when we condition our

[29] Note that our specification differs somewhat from the standard setting in which price, rather than degree of surge, would be the right-hand-side variable. Since surge is proportional to the fare absent surge, the longer the trip, the greater the dollar increase in the fare as surge increases. There are also differences in base fares per mile across cities in the sample. Thus, the elasticity estimates we report here represent a mixture across these various types of trips. We explore the impact on elasticities and consumer surplus of treating cities as distinct markets below.

[30] The particular groupings of the elasticity estimates do not materially impact our conclusions.

[31] As noted earlier, because it is extremely labor intensive to retrieve the continuous measure of expected wait time, we were only able to access two weeks of data. This continuous measure of wait time is our instrument for the discrete version that is seen by customers. Consequently, we estimate the first stage equation for expected wait times only in that subset of the data and then impose that coefficient on the full sample using a two-step procedure. Columns 4 and 5 thus are estimated using all of the observations in the data.

[32] Table A1 presents the number of observations and the R2 for each elasticity being estimated from each of the five regression specifications.

estimating equations on a particular consumer signaling an interest in making a purchase implies that these demand shocks are operating primarily on the extensive margin.[33] A second pattern is that there is little systematic change in the coefficients as controls are introduced or from instrumenting for expected wait time. At low surge levels (e.g., up to 1.5x), estimates that use all the variation in the data (column 1) are on average slightly smaller in absolute magnitude than those identified only off of the price discontinuities (column 1 versus column 2). Controlling for expected wait time also slightly increases the estimates (column 3 versus column 2). Instrumenting for wait time and adding other controls (columns 4 and 5) has little impact on the coefficients compared with RD estimates that add expected wait time as a control (Column 3). The third striking pattern in the table is the degree to which demand is price *inelastic* along the length of the demand curve. Of the fifty price elasticities estimated in the table, only five carry a coefficient greater than one in absolute value. The median point estimate is -0.54; the mean point estimate is -0.63. The fourth pattern in the table is that elasticities are quite precisely estimated at low prices, but become less precise at higher prices, because there are many more observations at low prices and percent price changes are much larger.

Table 5 explores how the estimated price elasticities vary across subsets of the data. To facilitate comparisons across subgroups, we compute a single price elasticity estimate that is an inverse variance-weighted average of the estimates reported in Column 4 of Table 4. The top row of Table 5 reflects the whole sample (i.e. all the data in Table 4), for which we find a demand elasticity of -0.72 (se=0.02). The next four rows of the table report elasticities separately for the four cities in our sample. Interestingly, demand is estimated to be substantially less elastic in Los Angeles (-0.49, se=0.05) than in the other three cities. The next eight rows of the table partition the hours of the week into eight mutually exclusive and collectively exhaustive categories.[34] We observe little variation in price elasticity along this dimension. Demand is estimated to be most elastic during the evening on weekends (-0.86, se=0.05) and least elastic during the evening weekday rush (-0.60, se=0.07). When we divide consumers by prior experience with Uber, the more frequent users (e.g., those taking more than 8 rides in a period) are the most elastic.

Sessions versus trips. Our outcome is whether a session ends in a request. A rider who declines a surge price may request an UberX ride shortly after the session ends, or switch to another Uber product such as UberPool, and our baseline treats both as lost purchases. Our estimates therefore measure the price elasticity of session conversion, which can exceed the price elasticity of trips taken. For consumer surplus from UberX, switching to another product is a genuine outside

[33] See the model in section 3 for a more formal discussion of why eliminating this source of error may be important for estimating demand in the context of our model

[34] The details of this time grouping are described in the results Appendix.

option and is correctly counted as a lost UberX purchase. Rebooking UberX a few minutes later is not: such riders still obtain an UberX trip, and counting them as lost purchases makes demand appear more elastic than it is. To the extent this occurs, it biases our elasticities away from zero and our consumer surplus estimates downward, so our estimates are conservative in this respect.

5. Turning Price Elasticities into Consumer Surplus Estimates

Armed with price elasticity estimates at various points along the demand curve, in this section we describe our methodology for mapping these into consumer surplus. We first carry out that mapping and ignore the fact that the estimates provided by the data are not precisely the estimates we require. This gives us a baseline estimate that is potentially biased. We then consider three potential problems with our approach, present approaches to dealing with those problems, and make conjectures regarding the likely sign and size of the bias caused by these data shortcomings.

5.1. Measuring Consumer Surplus

Transforming price elasticity estimates into an overall measure of consumer surplus requires additional assumptions. A thought experiment is helpful in this regard. Consider first the set of transactions that took place with no surge. To estimate consumer surplus via price changes, we would want to offer a menu of prices, identifying the price at which the consumer becomes indifferent to making the purchase, holding everything else constant. The sum of the differences between that willingness to pay and the price the consumer pays is the consumer surplus associated with those transactions. One would then carry out the same exercise for the transactions done at 1.2x surge, but using 1.2x as the base price, and so on, for each transacted price. The sum of these calculations would be total consumer surplus from UberX.[35]

What we actually observe in the data differs from this idealized thought experiment in at least three ways. First, we can only use our RD design to measure price elasticities at a handful of discrete points. Second, the price elasticities we see are derived from the population of sessions that appear near the threshold of the price change. That population of sessions is not necessarily the same as the population of sessions exposed to other prices. For instance, surge pricing is more common during rush hours and late at night; the willingness to pay for an Uber ride may vary systematically over the course of the day. Relatedly, if customers can predict surge and the most

[35] Note that there are sessions who are only offered, say, 1.5x surge by Uber, who do not transact at that price, but would have made a purchase had they been offered, say, 1.2x surge. These individuals are (rightly) ignored in the consumer surplus calculation because they didn't transact and thus received no consumer surplus from Uber. These individuals might, however, be of interest in the creation of hypothetical counterfactuals had Uber priced differently.

price-sensitive customers do not even open the app when they expect high surge, this could also bias the population of customers observed around the discontinuities. Third, in the thought experiment above, everything else is held constant as prices change. In reality, other demand determinants like the availability of taxis might differ between low surge and high surge situations.

We explore each of those potential problems below, but for the time being we start by simply assuming them away to generate a first pass estimate. For our first-pass estimates, with respect to a changing composition of sessions as surge changes and the possibility that outside options might also change, we begin by assuming those two concerns are unimportant, simply using the estimated elasticities from Table 4 and assuming that these elasticities are unbiased. To deal with discrete elasticity estimates, we follow the obvious path of assuming that the locally estimated elasticities also apply nearby. For instance, although our RD price elasticity going from 1.2x to 1.3x is identified off of variation from observations with generators between 1.24x and 1.26x, we will assume the same elasticity holds for sessions ranging from 1.20x to 1.30x. As noted above, we revisit these concerns below after explaining our basic methodology.

5.2. Estimates Using the Basic Methodology

Our basic methodology is straightforward. We start with the set of sessions who made a purchase at 1.0x surge. Assuming that the relevant price elasticity for this group at 1.2x surge is the price elasticity estimated using RD and reported in Table 4, we compute the number of sessions who would have continued to buy at 1.2x surge. Multiplying the price difference (20 percent) by the average fare actually paid ($13.3) multiplied by the quantity willing to pay 1.2x (in this case, 104MM trips) yields an estimate of the amount of consumer surplus generated up to the price 1.2x for those who paid 1.0x. Starting at that price and quantity, we then carry out an analogous calculation moving between 1.2x and 1.3x. That yields the consumer surplus for the same set of sessions, but associated with the consumer willingness to pay between 1.2x and 1.3x. We repeat that process until we reach 4.8x.[36] Beyond that point, where we have no elasticity estimates, we make the most conservative assumption, namely that no consumer is willing to pay above 4.9x.[37] Summing all of these surplus estimates yields the consumer surplus for sessions who faced 1.0x pricing. Figure 6 displays the outcome of this first step in the exercise visually.

[36] For consumer surplus, we use an inverse variance-weighted average of the less precise estimates created at high surge, as described in the elasticity section.

[37] If instead, we assume that the price elasticity associated with our highest level of surge holds beyond that point, our estimate of consumer surplus increases 34 percent.

We then repeat the exact same process outlined in the previous paragraph, but using sessions where a purchase was *actually* made at 1.2x (instead of 1.0x in the previous paragraph). Given the simplifying assumptions of our basic methodology, these sessions have the same price elasticities and thus an identically shaped demand curve, but they paid 1.2x, so their consumer surplus starts only at that point. The same approach is used for those who paid 1.3x, 1.4x, …, up to 4.9x.

Carrying out this exercise, we can extrapolate our estimates in these four cities from the first 24 weeks of 2015 using volume figures from all of 2015 to reach an overall consumer surplus estimate of $2.57 billion[38], with a bootstrapped standard error estimate of $141 million (which, as noted in Section 4.2, likely understates the true uncertainty).[39] These four cities accounted for about 42.6 percent of UberX U.S. gross bookings in 2015. If we assume that consumers elsewhere receive the same consumer surplus per dollar spent as consumers in our sample, so that consumer surplus is proportional to gross bookings, then our estimates imply that UberX generated $6.03 billion in consumer surplus in the United States in 2015. Surplus per dollar may differ in markets with different transit options and levels of competition, so this extrapolation should be read as a rough benchmark.[40] We now turn our attention to possible biases induced by the simplifying assumptions made above in estimating consumer surplus.

5.3. Sensitivity of the Results to Incorporating Continuous Elasticity Estimates

Our basic methodology above assumes a functional form for the elasticities that is linear with jumps at the RD points. A more nuanced approach predicts the elasticity at each point along the surge generator by fitting a curve using a flexible functional form. Doing this using a fifth-order polynomial yields consumer surplus estimates that are generally similar to those presented above. The two curves in Figure 7 are nearly identical, and the high order polynomial would imply an estimate of 4.7% less consumer surplus than our preferred method.

5.4. Sensitivity of the results to the possibility that the composition of the sessions changes over the surge distribution

[38] As noted in Table 5, different cities have different elasticities. If, instead of applying the blended elasticities to all fares, we apply each city's elasticities to the fares from that city and then sum the consumer surplus, we arrive at an estimate of $2.68B, implying $1.46 of consumer surplus per dollar of fare. A further analysis would be to do a similar sensitivity analysis grouping by rides with a similar fare, rather than by city, but this analysis is not feasible with current data availability.

[39] Table A2 in the online appendix breaks down this CS estimate for the different surge level multipliers.

[40] The different specifications in Table 4 do not produce meaningful differences in estimated consumer surplus.

As reported in Table 1 earlier in the paper, there are observable dimensions along which sessions systematically differ with the degree of surge. For example, the average surge at 1am on Saturday night is meaningfully higher than the average during the middle of the work day. The primary concern surrounding observable differences in the sample as a function of surge is that the price elasticities might vary along these dimensions.

We adopt three different approaches to deal with this issue. Our first approach is to use matching techniques to reweight the data so that the observable characteristics at a particular surge level match as closely as possible to the baseline population we are interested in. Say, for instance, we want to construct a counterfactual in which the set of sessions used to estimate the price elasticity at 1.5x mirrors the observable characteristics of the sessions that actually saw 1.0x surge.

To accomplish this matching on observables, we partition the data into 9,216 possible cells that represent the intersection of 144 geographic areas × 8 parts of the week × 4 measures of intensity of Uber use × 2 measures of whether the observation is before or after the calendar mid-point of our sample to capture possible changes over time given the rapid growth of Uber and the dynamic environment in which it operates. We observe sessions in 7,627 of these cells. We compute the share of sessions falling into each bucket b when surge is 1.0x (call this $S_{1.0x}$) and also the share in each of the buckets in the neighborhood of the 1.5x cutoff (call this $S_{1.5x}$), where b indexes the buckets. The resulting weight given to each observation in bucket b when estimating the price elasticity at the 1.5x cutoff is $S_{1.0x}/S_{1.5x}$. This ensures that the weighted number of observations per bucket in the vicinity of the 1.5 cutoff matches the number of observations in that same bucket at 1.0x surge.[41] Figure 8 provides a comparison of the estimated demand curves from our basic methodology and those using the reweighting technique. Until surge reaches 1.5x, the two curves look quite similar. Beyond that point, the re-weighted demand curve is substantially steeper. This implies that users who face high surge, but who look like no-surge users, are more likely to convert than the typical high-surge user. These results suggest that our basic approach understates consumer surplus by 23 percent (about $3.16 billion rather than $2.57 billion). Because the reweighting targets exactly the population that generates most of our estimated surplus, namely sessions quoted 1.0x, we view this as a central estimate rather than merely a robustness check; Section 5.8 discusses how we weigh it against the baseline.

Our second approach to addressing the problem that sessions that see high surge may systematically differ from those with low surge exploits the fact that pricing is a function of both

[41] A complication arises, of course, when there are no observations in this example in a particular bucket around the 1.5x threshold. Our baseline procedure in this scenario is to estimate the elasticity without this bucket, which assumes that sessions in this bucket would have behaved like the sessions we actually observe near the 1.5x threshold.

demand and supply conditions. Even when levels of demand are completely ordinary, consumers can face high prices if supply is unusually low. Price changes caused by a shift in supply holding demand constant are likely to provide a more compelling counterfactual for constructing consumer surplus estimates than those driven by spikes in demand.[42] We estimate separate regressions of the form below in data where each observation is a location in a particular five minute time period:

$$Sessions_i = \alpha + \beta(Week\ of\ year)_i + \gamma(Hour\ of\ week)_i + \epsilon \quad (5)$$

where the dependent variable is the number of sessions initiated by consumers within a given five minute time period *t* in one of the 783 mutually exclusive and exhaustive geographic units in our sample.[43] On the right hand side, we include as indicators the week of the year and the hour of the week (24 hours x 7 days per week). The fitted value of this regression is the expected number of consumer sessions originating in a particular time and place. Under the hypothesis that price elasticities may systematically differ during demand spikes, we then explore two different subsets of the data. First, we exclude any observation where the observed number of sessions was greater than the number of sessions our model predicted.[44] Second, we exclude observations in the top and bottom quartile of demand shocks, i.e. we keep only the middle half of the data. As would be expected, Table 6 shows that high surge activity is greatly reduced in these two subsets. Compared to the full data set, the mean level of surge in our two subsets is about half as high (1.14 versus roughly 1.07). The effects are most extreme for high surge levels: our two subsets each comprise about half of the total observations, but capture less than 20 percent of the surges above 2.0 and less than 10 percent of surges above 3.0. It is rare, empirically, to achieve high surge levels based on supply shocks alone.

Figure 9 compares our baseline demand estimates to those derived using these two subsets of the data. Up until a surge of roughly 2.0x, the three curves look nearly identical. Beyond that point, but especially at the very highest surge levels, the estimates based on the subsets become extremely noisy. Because the subset estimates are too noisy at high surge to support a reliable

[42] If the RD approach we use is valid, then demand shocks are a useful form of variation for identifying price elasticities. The concern here is not the usual one of endogeneity. Rather, the concern is that during demand spikes consumer preferences differ from typical demand periods. In other words, using demand shocks we can get consistent estimates of the price sensitivity for the population facing those prices; that population may not, however, be representative of the typical consumer.

[43] We treat each update to the geography of a unit as a separate unit. The number of geographic units at a particular time is substantially smaller.

[44] Exploration of the most extreme positive demand shocks confirmed the unusual nature of the circumstances, e.g. rainstorms so severe that widespread flooding occurred, major concert and sporting events, etc.

consumer surplus figure, we do not report one; over the range where they are precise, they are nearly identical to the baseline.

Our third check on whether high-surge sessions differ from low-surge sessions explores whether the decision to open the Uber app is sensitive to predicted levels of surge. In particular, if price sensitive customers disproportionately choose not to open the app when they expect surge (and those forecasts are accurate), then price insensitive customers will be overrepresented when surge is predictably high, leading to elasticity estimates that are biased toward zero.[45] We test this hypothesis empirically in two ways. First, we see whether our elasticity estimates systematically differ across periods of *predictably* high surge and *unpredictably* high surge – only when surge is high *and* predictable would one expect customers to avoid opening the app. We use a regression similar to the regression that predicts sessions, but instead use the level of surge as the outcome:

$$Surge\ Multiplier_i = \alpha + \beta(Week\ of\ year)_i + \gamma(Hour\ of\ week)_i + \epsilon \qquad (6)$$

Using this simple model, we partition the population into observations either above or below the median predicted surge, re-estimate the elasticities, and plot the chained demand function.

Areas with low expected surge rarely surge, so the estimates tend to be less well estimated for the low-expected surge, particularly above 1.8. When we use an inverse variance-weighted average of the estimates from the 1.75 discontinuity and below, the high expected surge estimates have a mean of -.735 and a SD of .0129, and the low-expected surge have an elasticity of -.798 with SD of .0335. This difference is not significant, though a higher elasticity when surge is expected to be low is directionally consistent with a small amount of selection associated with non-opening of the app in anticipation of high surge. To gauge the size of any such bias, note that the low-expected-surge elasticities are about 9 percent larger in absolute value than the high-expected-surge elasticities. Under a constant-elasticity approximation to our demand curve with an elasticity between -0.6 and -0.7 and no willingness to pay above 4.9x, scaling all elasticities up by 9 percent reduces consumer surplus by roughly 4 to 5 percent. This provides an approximate upper bound on the bias from selective app opening, since it attributes the entire (statistically insignificant) difference to selection.

In Figure 11, we plot the demand curves implied by the two populations. For low surge values, they are very similar and diverge as the low expected surge populations lose power at high surge levels. Overall, this approach suggests that while it is possible some small amount of

[45] If all customers are equally discouraged by the threat of surge pricing, no systematic bias is induced in our estimates.

selection happens on the dimension of whether or not to open the app, any effect is small enough that it cannot be confidently identified using our data. We do not see any evidence that this type of selection materially changes our conclusions.

Our second approach examines the decision to open the app more directly. Using sessions through March 31, 2015, we classify as high-elasticity riders those who were quoted a surge of 1.5x or higher at least four times and never requested a ride at those prices. We then track, for the remainder of the period, the share of these riders in the population of sessions at different surge levels. If price-sensitive riders avoided opening the app when surge was high, their share would decline as surge rises. As Figure 12 shows, we do not observe such a decline. This test has a limitation: it compares the composition of riders who did open the app at different realized surge levels, so it detects avoidance only insofar as it shows up as a lower share at high surge. A rider who avoids the app entirely when surge is predictable would not be captured. The test is therefore best read as complementary to the predictable-surge comparison above rather than as a stand-alone test.

Qualitatively, it is not surprising that we see little evidence of selective app opening given the costs and benefits involved. It takes a consumer only tens of seconds to open the app and determine the actual level of surge. Given that we estimate the typical trip using Uber generates over $10 of consumer surplus, even a small chance of a normally priced trip should justify the exceptionally small cost when the consumer is in the market for a ride.

5.5. Sensitivity of the results to the possibility that other (mostly unobservable) demand determinants covary with the degree of surge

The assumption underlying a demand curve is that all else is held constant as the price of the good in question changes. We do not, in our setting, have good measures of the other demand determinants, such as the availability of taxi cabs, how hard it is raining, whether or not the person has an umbrella, or how costly it will be to the consumer if he or she get to their location ten minutes late. Most likely, the paucity of outside options will make demand for Uber more inelastic. If high surge is correlated with bad outside options for users, then biased elasticity estimates will likely lead us to overstate consumer surplus.[46]

[46] It is possible, also, that the number and type of consumers who open the Uber app will be a function of outside options, further complicating this analysis. If Uber users on the extensive margin are more price sensitive, and those users disproportionately appear when outside options are bad and surge is high, then the empirical bias could actually go in the other direction.

Although we cannot deal directly with this issue because we do not observe outside options in our data, we are able to address this potential confound in three ways. The first approach is the standard one -- to control for an increasing number of dummies and interactions and to observe how this affects the estimates. Additional controls had little impact on the coefficients in Table 4. Moreover, when we include dummies for the nearly 8,000 partitions described above, the estimated elasticities do not systematically change.

The specific structure of the Uber surge algorithm and business rules provide two other ways to shed light on the possible biases induced by unobservable factors, which are very different from the approach of adding more and more controls. The first of these exploits the continuous nature of Uber's surge algorithm combined with the discontinuous pricing jumps. The other exploits a business rule imposed by Uber that leads a subset of sessions to face prices that differ sharply from what the surge algorithm judges to be optimal. We deal with these two different approaches in turn.

5.6. Do users who experience worse market conditions, but the same prices, have different purchase likelihoods?

Market conditions differ across consumers who are quoted the same price. Consumers quoted 1.3x surge have underlying surge generator values ranging from 1.25 to 1.35. To the extent that outside options are worse on average at high surge relative to low surge, that same pattern should be evident within these more narrow windows as well: the sessions at 1.35 should have worse outside options than those at 1.25. And if this is the case, then one would expect a higher share of purchases by those at 1.35 than for those at 1.25. The key to this analysis is that we see consumers with known differences in underlying supply/demand conditions, but who all face the same price. In typical market settings, we do not get to observe this sort of variation, because prices equilibrate supply and demand at each point. It arises in this case only because Uber holds prices fixed over a range of market conditions. If prices are not held constant, we lose our ability to isolate the influence of outside options.

To empirically test the concept in the preceding paragraph, we run regressions with an indicator for whether a purchase is made as the dependent variable, with indicators for each price level on the right hand side, along with Uber's continuous measure of surge. Because of the inclusion of the price-level indicators, the identification of the continuous measure comes only

from consumers who face different market conditions, but the same price.[47,48] The estimated coefficient on the continuous surge measure is 0.0104 (SE=0.00). So, indeed, by this metric, it appears that outside options are slightly worse at higher levels of surge. The magnitude of the effect is quite small. The degradation of outside options moving from 1.0x to 3.0x would lead purchase rates, holding actual price constant, to increase by 2.08 percent, implying that our consumer surplus estimates above may be biased upward by about 3 to 4 percent.

5.7. Do individuals presented with prices well below what is implied by demand conditions behave differently than others facing those prices?

In the discussion above, we exploited relatively small differences in demand conditions across sessions offered the same price. Uber business rules generate a different natural experiment that leads, on occasion, to consumers who face very different supply/demand conditions to face the same price. Specifically, Uber limits the extent to which each consecutive update to the price can raise the surge price. At the beginning of a sharp demand/supply imbalance, the surge price can only increase by .5 in the first increase, .6 in the second increase, etc. As a consequence, there are cases in which sessions see a price of 1.5x, but the surge generator characterizes the market setting as warranting a surge price of 2x, 3x, or even higher. If the outside options at higher surge are worse, then we would expect that purchase rates of sessions facing high surge conditions *but artificially low prices of 1.5x* will be higher than the purchase rates of sessions who see prices of 1.5x because that is what market conditions warrant.

Figure 13 shows virtually no relationship between underlying demand conditions and purchase rates in those cases where price is artificially restricted by internal business rules. While cases where the underlying model is changing prices rapidly may not perfectly represent standard scenarios, this does suggest that there is little or no bias from this channel in the consumer surplus estimates above.[49]

5.8. Summary: Consumer Surplus under Alternative Assumptions

[47] Essentially this regression tells us whether, in Figure 5, there is a systematic trend in the height of the gray bars that fall within each pair of red and yellow bars.

[48] For computational feasibility, this regression is estimated using all sessions with surge > 1 and 15% of sessions where surge = 1. For specification, see Appendix: Within surge level estimator specification.

[49] One caveat on this conclusion is that many more riders (relatively to available supply) are opening the app for high generators, causing higher ETAs. However, from 1.5x to 4.0x, a 166% increase in the estimated market price, ETAs only increase by ~26%. Given the coefficients on ETA in purchase regressions, this change in ETA would not have dramatic effects on purchase rates.

Table 7 collects the consumer surplus implied by each of the alternative assumptions examined in this section. Where a section reports only a percentage change, we apply it to the baseline estimate of $2.57 billion. The last column divides by total UberX spending in the four cities (about $1.84 billion), and the U.S. column scales each estimate by the four cities' 42.6 percent share of U.S. gross bookings.

Three points stand out. First, the estimates range from roughly $2.4 billion to $3.4 billion, and every alternative we can compute implies consumer surplus of at least $1.30 per dollar spent. Second, the two largest adjustments both concern the upper part of the demand curve: how representative high-surge sessions are of base-price riders, and what is assumed above the highest multiplier we observe. Third, because about 84 percent of our surplus comes from sessions that paid 1.0x (Table A2), the estimate depends chiefly on whether sessions near the surge thresholds share the willingness to pay of base-price riders. The reweighted estimate addresses this directly for observable characteristics, and on that basis we regard $3.16 billion as at least as credible as the baseline. We retain $2.57 billion as our headline figure because it relies on fewer modeling choices and is conservative relative to the reweighted estimate.

6. Conclusion

This paper exploits the remarkable richness of Uber data to investigate the impact that Uber's introduction has had on consumer welfare. Our approach exploits the fact that Uber (1) has detailed session-level data, even when no purchase is made, (2) varies prices with market conditions, and (3) has business rules that generate sharp discontinuities in the prices that similar customers face. We find that consumer demand is inelastic, despite the existence of what would seem to be reasonably close substitutes (competitors, taxis, public transportation, driving one's self). Inelastic demand translates into large consumer surplus estimates: roughly $2.57 billion dollars in 2015 for the four cities in our sample under our baseline assumptions (about $3.16 billion when we reweight to match base-price riders), or $6.03 billion if extrapolated to all UberX trips in the U.S. for that year. This estimate of consumer surplus is about 1.9 times the revenues received by driver-partners and at least 5.6 times the revenue captured by Uber after the driver-partner's share is removed.

Several features of the market could account for the size of this surplus relative to spending: lower search and matching costs for riders relative to street-hail taxis, shorter and more predictable wait times, the ability to summon a car in areas where taxis rarely circulate, and cashless payment. Some of the surplus may also reflect lower prices relative to regulated taxi fares, part of which is

a transfer from medallion holders rather than new value. Our data do not allow us to separate these channels, and we leave their decomposition to future work.

The consumer surplus estimates we generate correspond to a short-run demand curve, because they are identified from transient price shocks while all other transportation options, including other Uber products and competing ride-share services, are held fixed. One day's worth of consumer surplus from UberX in our four cities is about $7.0 million, or about $16.5 million nationally on our extrapolation. This is roughly what consumers would lose if UberX alone were unexpectedly unavailable for a day. Mapping this number to other counterfactuals requires care, and the direction of the adjustment depends on the question. If UberX disappeared permanently while competitors remained, consumers would adjust over time and competitors would expand, so long-run demand for UberX would be more elastic and the surplus loss smaller than our annualized figure suggests. A ban on ride-sharing is a different counterfactual: it removes UberX together with its closest substitutes. The relevant quantity is then the combined surplus from all ride-sharing products, not the surplus from UberX computed with a steeper demand curve. Our estimate is one component of that total, and because consumers who lose UberX today can turn to other ride-sharing products, it likely understates the loss from removing ride-sharing as a whole. Estimating that total would require demand estimates for the full set of ride-sharing products, which our data do not cover.

One feature of our study may be of particular interest to the field of demand estimation: the previously unrecognized value of a data source in which individuals signal interest in a purchase prior to learning the market price. Our aggregate market estimates of demand had the wrong sign, but our estimated elasticities using individual data with simple OLS, conditioned on a person opening the app, were similar to our RD estimates. Because the market-level and individual-level estimates also differ in their level of aggregation and source of variation, our results do not isolate the contribution of conditioning alone; they are, however, consistent with conditioning on interest in making a purchase removing an important source of error in demand estimation. Re-estimating the market-level regression with the conversion rate (requests per session opened) as the dependent variable would isolate this contribution directly. We conjecture that this finding could generalize to a growing number of markets.

The emphasis in demand estimation over the last two decades has been on methodological advances that allow researchers to overcome the inherent limitations in the sorts of data that have typically been available. While recognizing the immense contributions of that work, this paper also points to a second path forward: one in which better data are the key to deeper insights. Massive changes that are taking place in the economy in terms of the availability of transaction-

level data, the increased use of sophisticated pricing tools by firms, and the growing openness of firms to randomized experiments. All of these forces point towards a future world in which data richness transforms our understanding of firms and consumers.

References

Ashraf, Nava, James Berry, and Jesse M Shapiro. 2010. Can Higher Prices Stimulate Product Use? Evidence from a Field Experiment in Zambia. *American Economic Review* 100(5): 2383-2413.

Baker, Jonathan B. and Timothy F. Bresnahan. 1988. "Estimating the residual demand curve facing a single firm." *International Journal of Industrial Organization* 6(3): 283-300.

Berry, Steven and Philip Haile. 2014. "Identification in Differentiated Products Markets Using Market Level Data." *Econometrica* 82(5): 1749-1797.

Berry, Steven, James Levison, and Ariel Pakes. 1995. "Automobile Prices in Market Equilibrium." *Econometrica* 60(4): 889-917.

Bresnahan, Timothy. 1986. "Measuring the Spillovers from Technical Advance: Mainframe Computers in Financial Services." *American Economic Review* 76(4): 742-755.

Buchholz, Nicholas. 2016. "Spatial Equilibrium, Search Frictions, and Efficient Regulation in the Taxi Industry." (Job market paper, UT Austin).

Camerer, Colin F., Linda Babcock, George Loewenstein and Richard Thaler. 1997. "Labor supply of New York City cab drivers: One day at a time." *Quarterly Journal of Economics* 112(2), 408-441.

Castillo, Juan C. 2025. Who benefits from surge pricing?. *Econometrica*, 93(5), pp.1811-1854.

Christensen, Peter, Gustavo Nino, and Adam Osman. 2023. *The demand for mobility: Evidence from an experiment with uber riders* (No. w31330). National Bureau of Economic Research.

Cohen, Jessica and Pascaline Dupas. 2010. Free Distribution or Cost-Sharing? Evidence from a Randomized Malaria Prevention Experiment. *Quarterly Journal of Economics* 125(1): 1-45.

Cramer, Judd. 2016. "The Effect of Uber on the Wages of Taxi and Limo drivers." (Job market paper: Princeton University).

Crawford, Gregory S., Robin S. Lee, Michael D. Whinston, and Ali Yurukoglu. 2018. "The Welfare Effects of Vertical Integration in Multichannel Television Markets." *Econometrica* 86 (3): 891-954.

Crawford, Vincent P. and Juanjuan Meng. 2011. "New York City Cab Drivers' Labor Supply Revisited: Reference-Dependent Preferences with Rational Expectations Targets for Hours and Income." *American Economic Review* 101(5): 1912-32.

Crawford, Gregory S. and Ali Yurukoglu. 2012. "The Welfare Effects of Bundling in Multichannel Television Markets". *American Economic Review* 102(2): 643-85.

Cullen, Zoë and Chiara Farronato. 2014. "Outsourcing Tasks Online: Matching Supply and Demand on Peer-to-Peer Internet Platforms." (Job market paper: Stanford University).

Deaton, Angus. 1986. "Demand analysis." *Handbook of Econometrics* 3: 1767-1839.

Dupas, Pascaline. 2014. Short-Run Subsidies and Long-Run Adoption of New Health Products: Evidence From a Field Experiment. *Econometrica* 82(1): 197-228.

Einav, Liran, Chiara Farronato, and Jonathan Levin. 2016. "Peer-to-Peer Markets." *Annual Review of Economics* 8: 615-635.

Einav, Liran and Jonathan Levin. 2010. Empirical Industrial Organization: A Progress Report. *Journal of Economic Perspectives* 24(2): 145-162.

Farber, Henry. 2005. "Is Tomorrow Another Day? The Labor Supply of New York City Cab Drivers." *Journal of Political Economy* 113(1): 46-82.

Farber, Henry. 2008. "Reference Dependent Preferences and Labor Supply: The Case of New York City Taxi Drivers." *American Economic Review* 98(3): 1069-1082.

Farber, Henry. 2015. "Why You Can't Find a Taxi in the Rain and Other Labor Supply Lessons from Cab Drivers." *Quarterly Journal of Economics* 130(4): 1975-2026.

Farronato, Chiara and Andrey Fradkin. 2018. The Welfare Effects of Peer Entry in the Accommodation Market: The Case of Airbnb. Working Paper.

Fraiberger, Samuel and Arun Sundararjan. 2015. "Peer-to-Peer Rental Markets in the Sharing Economy." NYU Stern School of Business Research Paper.

Goldszmidt, Ariel, John A. List, Robert D. Metcalfe, Ian Muir, V. Kerry Smith, and Jenny Wang. 2020. The Value of Time in the United States: Estimates from Nationwide Natural Field Experiments. National Bureau of Economic Research, No. w28208.

Goolsbee, Austan and Amil Petrin. 2004. Gains from Direct Broadcast Satellites and the Competition with Cable TV." *Econometrica* 72(2): 351-381.

Hahn, Robert and Robert Metcalfe. 2021. Efficiency and Equity Impacts of Energy Subsidies. *American Economic Review*.

Hall, Jonathan, Cory Kendrick, and Chris Nosko. 2015. "The Effects of Uber's Surge Pricing: A Case Study". unpublished.

Hall, Jonathan V. and Alan B. Krueger. 2015. "An Analysis of the Labor Market for Uber's Driver-Partners in the United States." Working paper #587, Princeton University Industrial Relations Section.

Hendel Igal, and Aviv Nevo. 2006. "Measuring the Implications of Sales and Consumer Inventory Behavior." *Econometrica* 74(6): 1637-1673.

Hortaçsu, Ali and Jakub Kastl. 2012. Valuing Dealers' Informational Advantage: A Study of Canadian Treasury Auctions. *Econometrica*, 80(6): 2511-2542.

Hortaçsu, Ali, Allen Zhang, and Jakub Kastl. 2018. Bid Shading and Bidder Surplus in the U.S. Treasury Auction System. *American Economic Review*, 108(1): 147-169.

Houghton, R.W. 1958. "A Note on the Early History of Consumer's Surplus." *Economica* 25(97): 49-57.

Jessoe, Katrina and David Rapson. 2014. Knowledge Is (Less) Power: Experimental Evidence from Residential Energy Use. *American Economic Review* 104(4): 1417-38.

Mortimer, Julie Holland. 2007. "Price discrimination, copyright law, and technological innovation: Evidence from the introduction of DVDs." *Quarterly Journal of Economics* 122(3): 1307-1350.

Nevo, Aviv. 2000. "Mergers with Differentiated Products: The Case of the Ready-to-Eat Cereal Industry." *RAND Journal of Economics* 31(3): 395-421.

Petrin, Amil. 2002. Quantifying the Benefits of New Products: The Case of the Minivan, *Journal of Political Economy* 110(2): 705-729.

Quan, Thomas W., and Kevin R. Williams. 2018. "Product Variety, Across Market Demand Heterogeneity, and the Value of Online Retail". *RAND Journal of Economics* 49(4): 877-913.

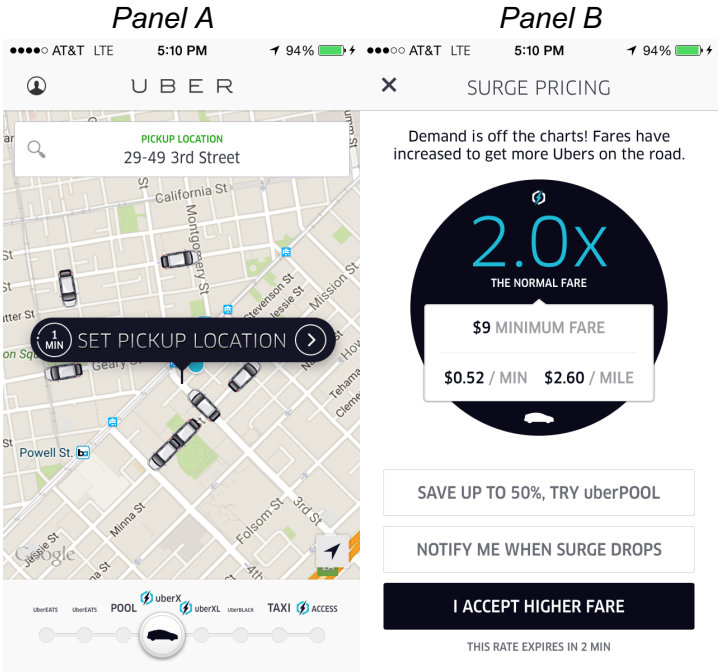
Svoboda, M. 2008. "History and Troubles of Consumer Surplus," *Prague Economic Papers* 17(3): 230-242.

Williamson, Oliver. 1968. "Economies as Antitrust Defense: The Welfare Trade-Offs." *American Economic Review* 58(1): 18-36.

Willig, Robert D. 1976. "Consumer's Surplus Without Apology." *American Economic Review*. 66(4): 589-597.

Figures

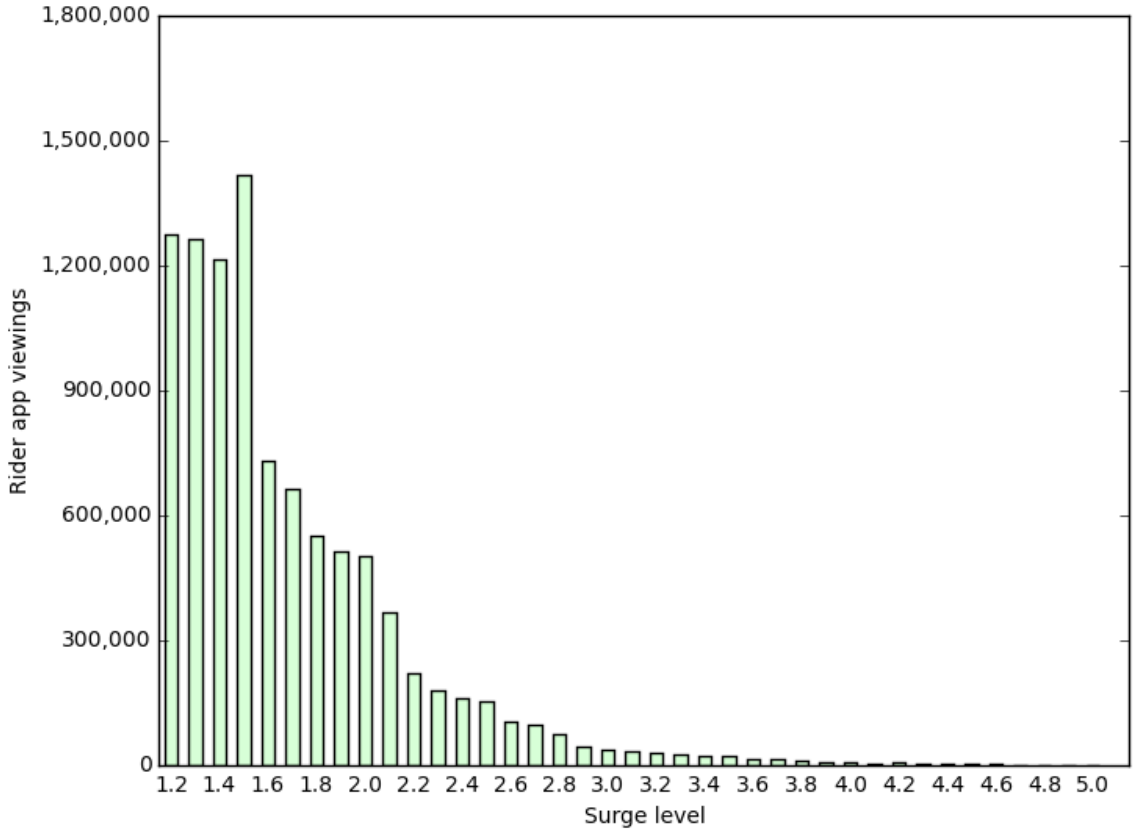
Figure 1: Uber mobile application request screens



Note: These figures illustrate what the Uber app looks like[50] when a rider is requesting transportation. Panel A depicts the period preceding a request when users are asked to choose a product and set a pick-up location. Panel B depicts the confirmation screen where users are presented with a surge price when applicable.

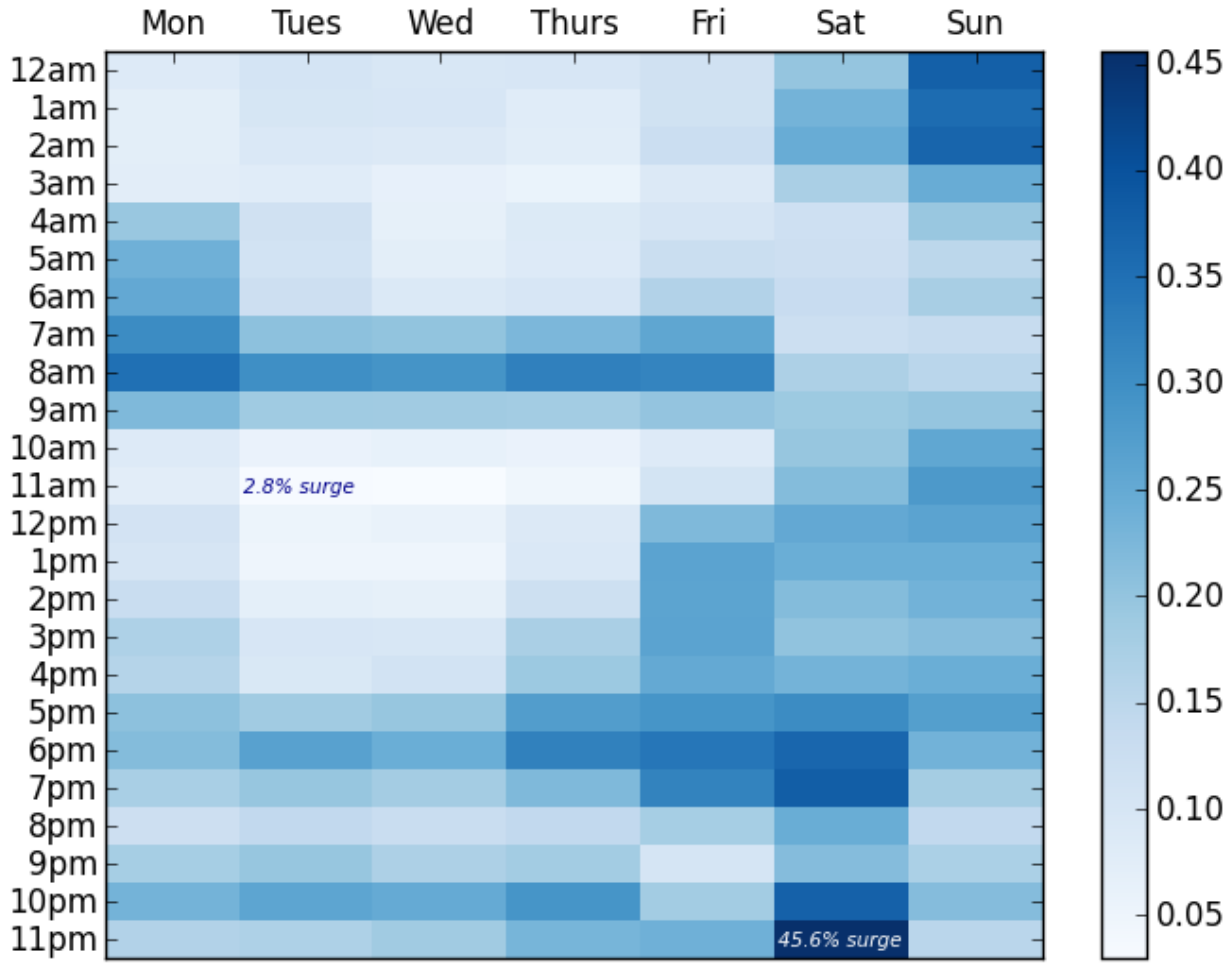
[50] These pictures are representative. In practice, there are variations across geography and time in terms of visual layout and exact product availability.

Figure 2: Distribution of surge price sessions for surge prices greater than 1.0x



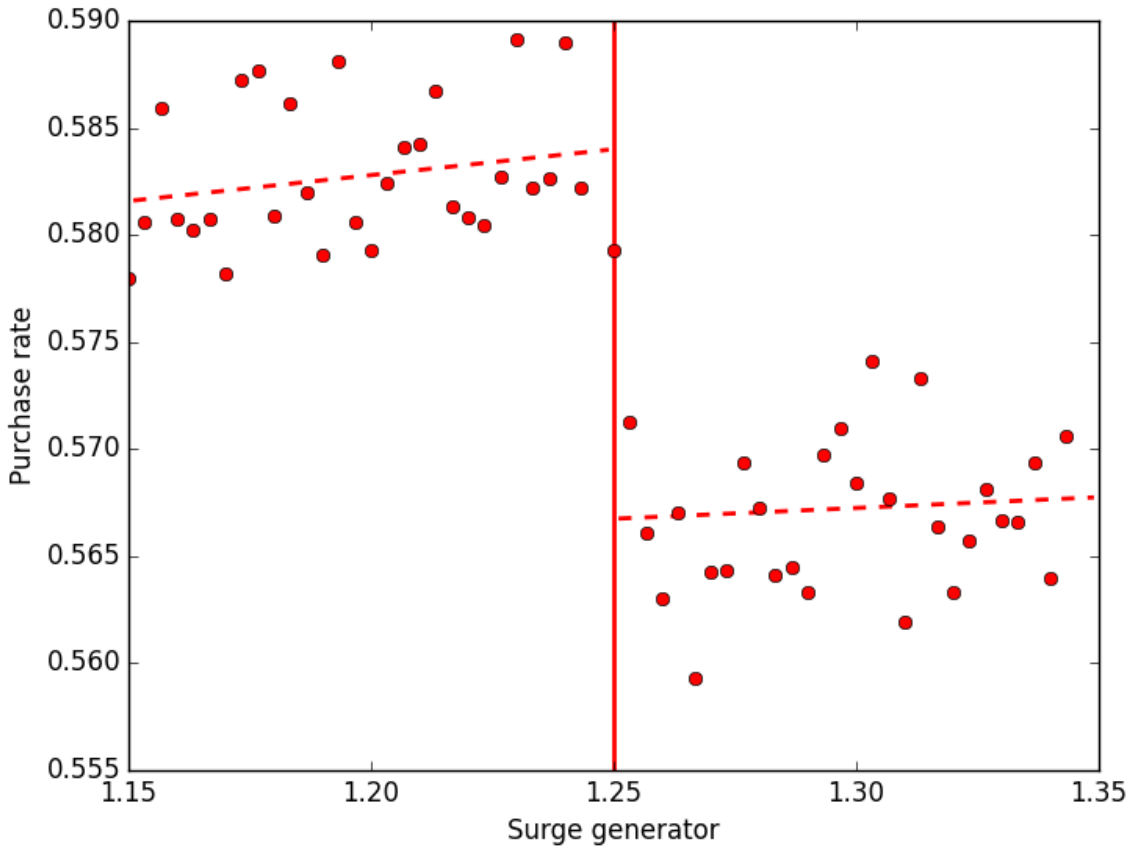
Note: This figure presents the number of observed UberX surge prices by surge level. Rides with no surge are excluded. Surge price notation is abbreviated. For example 1.2 in the graph corresponds to a surge price of 1.2x.

Figure 3: Heat map of the percent of sessions with surge by hour of the week



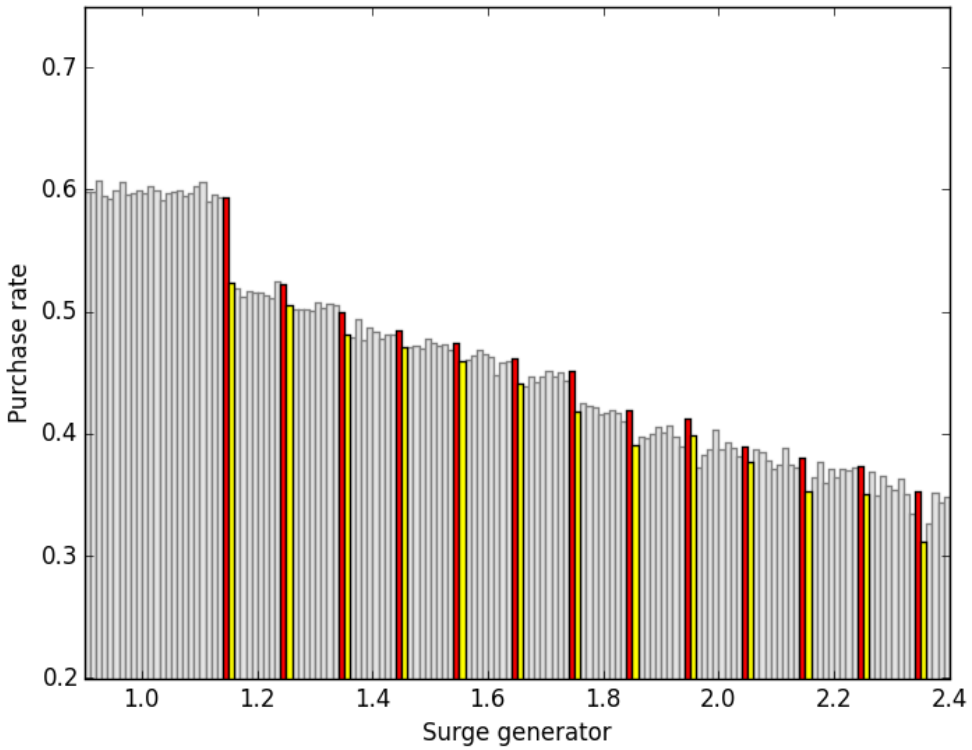
Note: This figure shows the frequency of surges by hour of day and day of week for UberX. Darker rectangles identify times and days when riders are more likely to face surge pricing. Tuesday at 11am represents the time and day combination when surge pricing is least common, and Saturday at 11pm represents the time and day combination when surge pricing is most common.

Figure 4: Example of purchase rate changes at price discontinuity



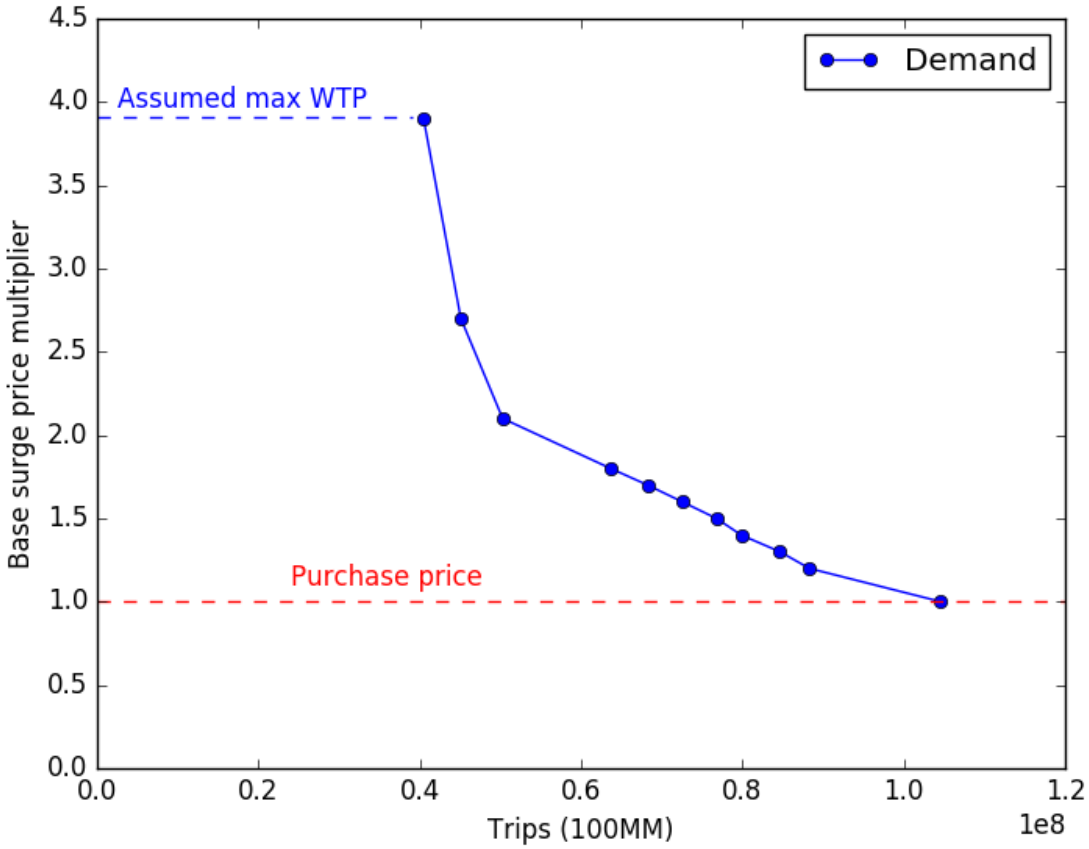
Note: This figure illustrates how purchase rates vary as a function of the surge generator over the range 1.15x to 1.35x. The vertical line when the surge generator equals 1.25 identifies the point at which the surge price changes from 1.2x to 1.3x.

Figure 5: Purchase rate drops at pricing discontinuities



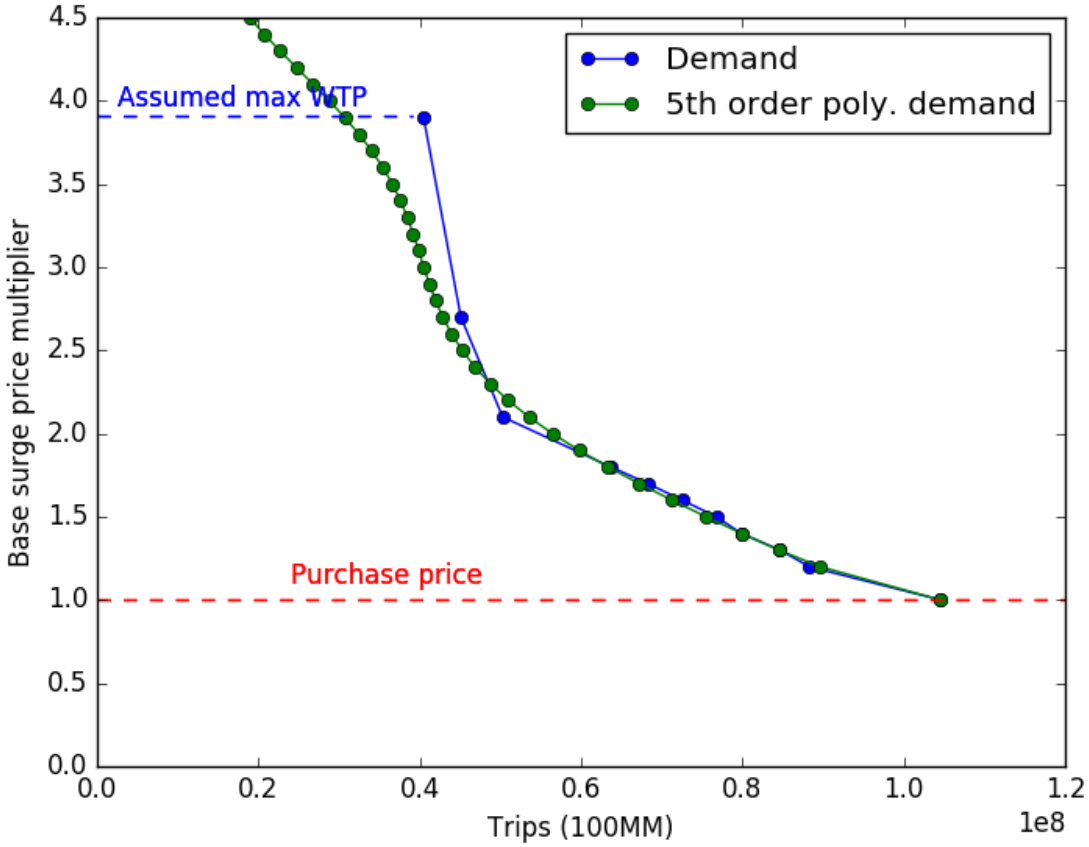
Note: This figure illustrates how purchase rates vary as a function of the surge generator when the surge generator is less than 2.4x. Red bars identify all observations within .01 units to the left of a price discontinuity. Yellow bars identify all observations within .01 units to the right of a price discontinuity. All observations not within these windows are depicted in gray.

Figure 6: Visual representation of demand curve for transactions at 1.0x



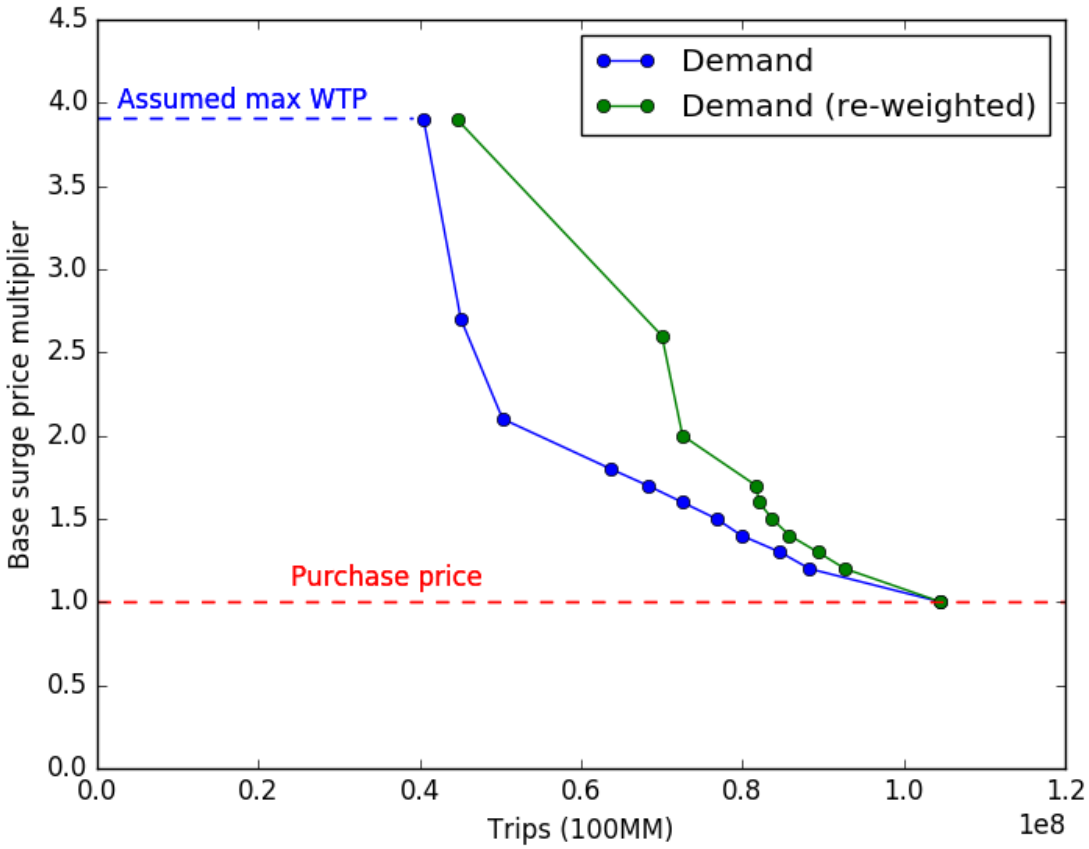
Note: This figure presents a piecewise linear demand curve with jumps at each price discontinuity. The curve is generated from the underlying elasticities estimated for each price discontinuity and for consumers facing transactions at 1.0x.

Figure 7: Comparison of 5th order smooth and linear segmentation demand curves



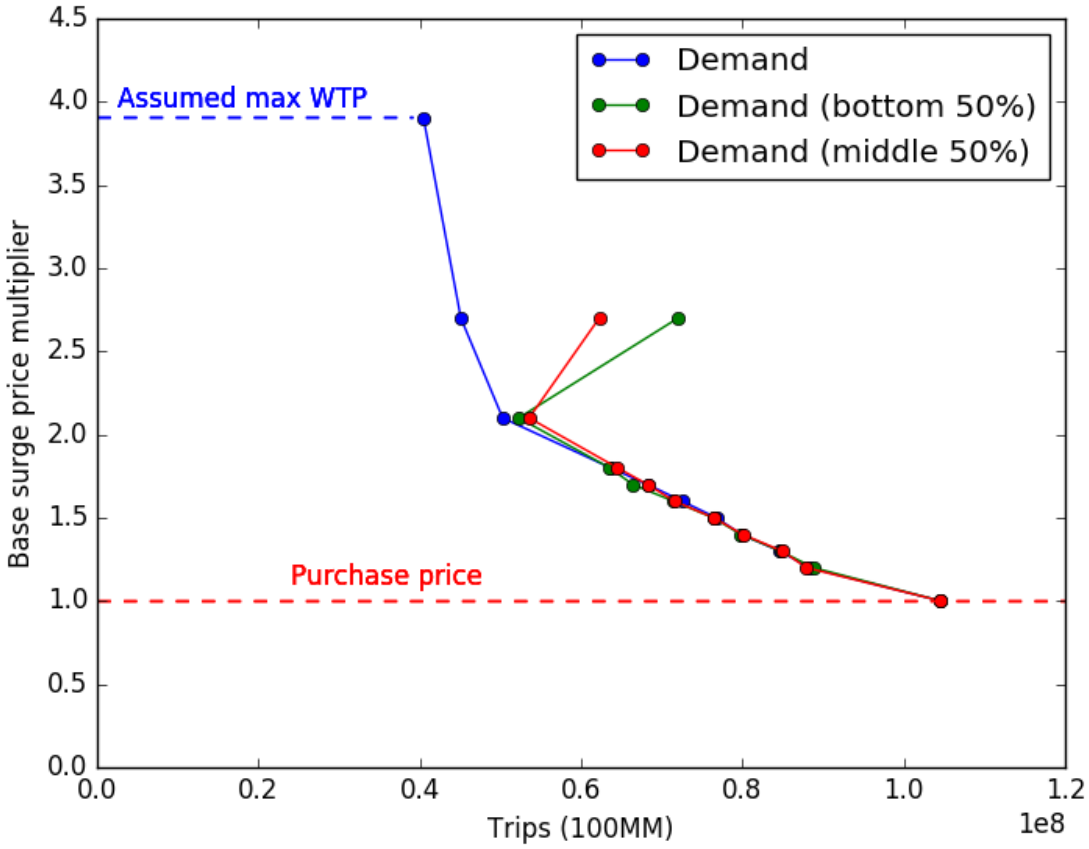
Note: This figure presents two demand curves generated via different approaches. The blue demand curve (also presented in Figure 6) is piecewise linear with jumps at each price discontinuity while the green demand curve is derived by fitting a 5th order polynomial to the elasticity estimates.

Figure 8: Elasticity estimates with and without matching on observables



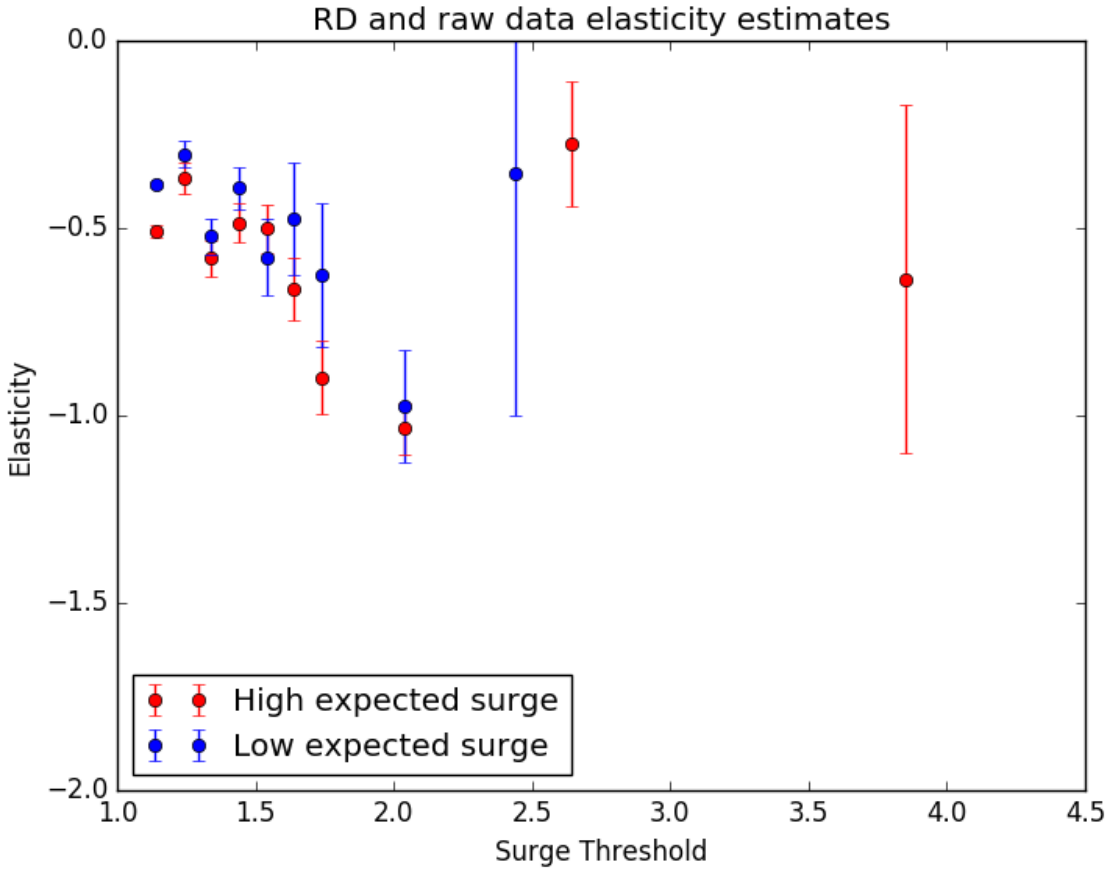
Note: This figure presents two demand curves generated via different approaches. The blue demand curve (also presented in Figures 6 and 7) is linear with jumps at each price discontinuity while the green demand curve is based on elasticity estimates derived from data that were re-weighted to match the distribution of observables found at price 1.0x.

Figure 9: Estimated demand curves with and without extreme demand shocks



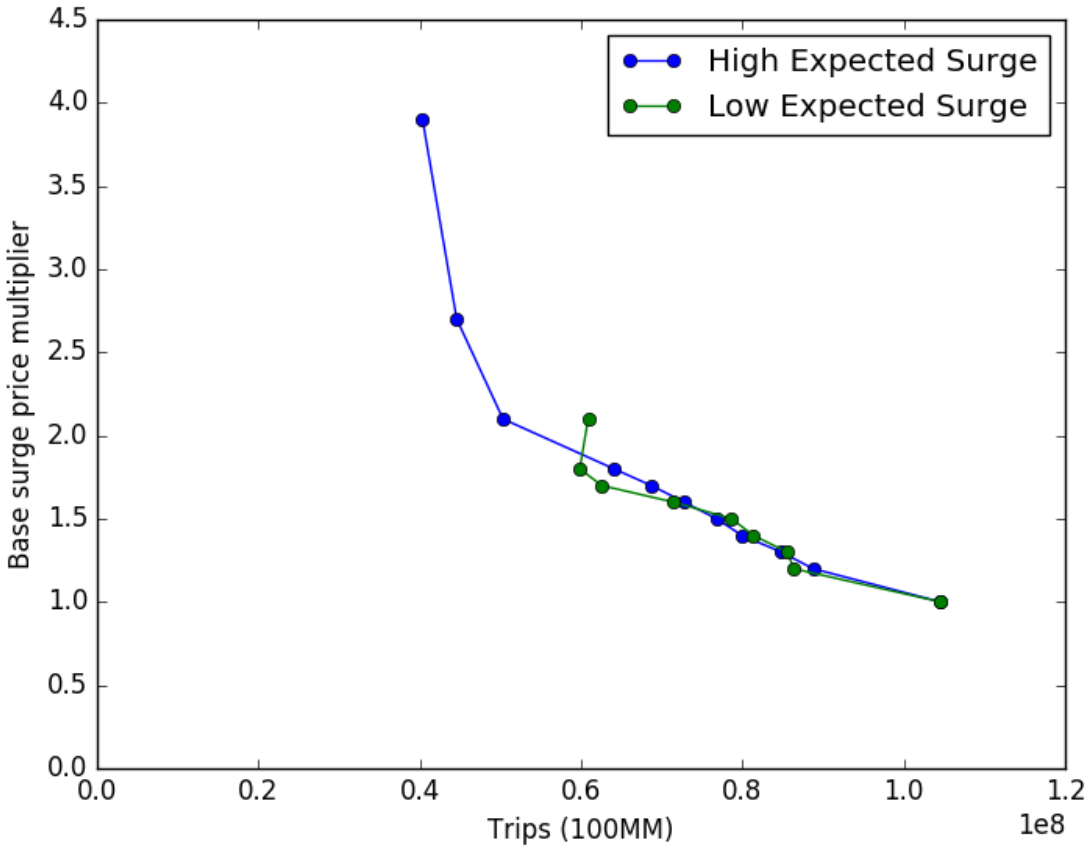
Note: This figure presents three demand curves generated via different approaches. The blue demand curve (also presented in Figures 6, 7, and 8) is linear with jumps at the price discontinuities. The green demand curve restricts the analysis to observations where demand is below the median level, while the red demand curve restricts the analysis to observations where demand is around the median level. We do not plot the highest demand curve point due to noisy estimates

Figure 10: Elasticity estimates by expectation of surge



Note: This figure represents elasticities for riders who had different expectations of surge, as defined by a simple model of when surge is consistently present.

Figure 11: Estimated demand curves by expected surge



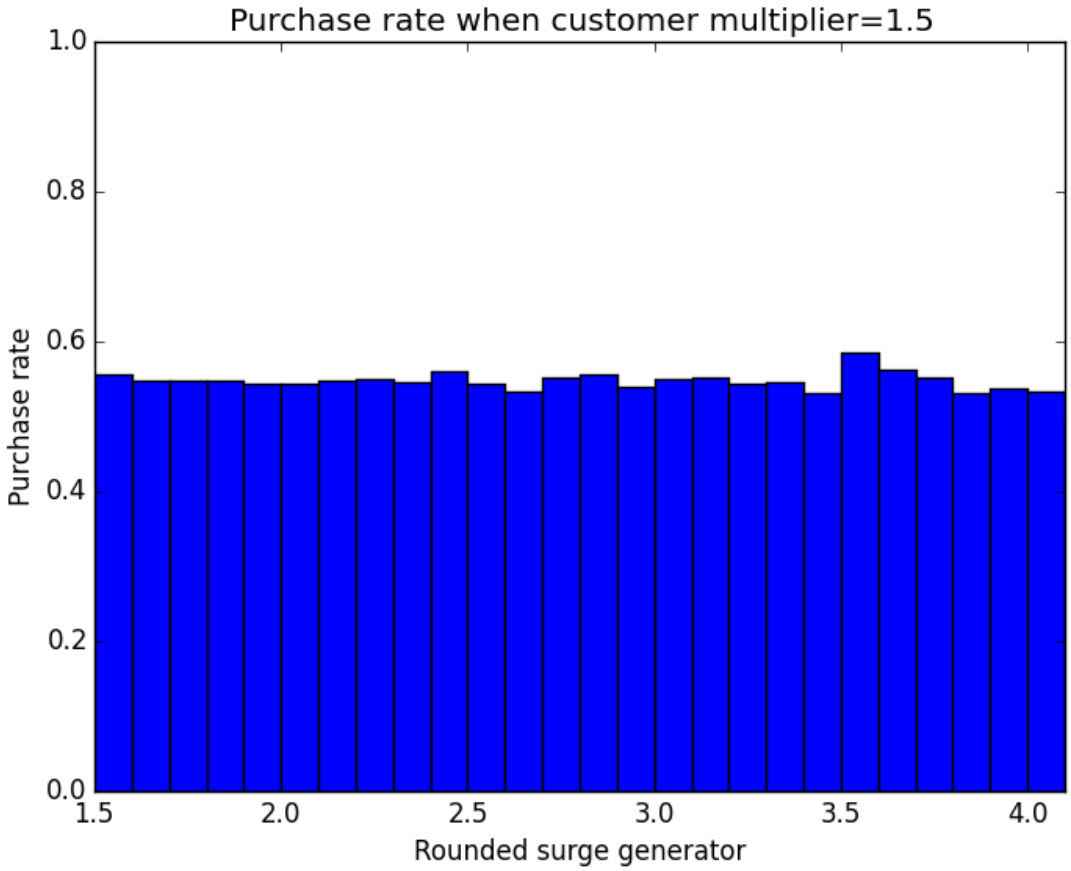
Note: The high expected surge represents observations where a simple model would predict surge to be high. Above our 2.1 elasticity estimates, the low expected surge elasticities are too poorly estimated to be meaningful.

Figure 12: High elasticity riders as a percentage the population seeing high surge, by surge



Note: In the first period of our sample (through March 31, 2015), we define riders as high elasticity if they were quoted a surge of 1.5x or higher at least four times and never requested a ride at those prices. We then chart, for the second period, these riders as a proportion of the population of all riders who saw at least four surges of this level. The lack of a downward trend indicates that high-elasticity riders do not become less common among app users as surge rises, which is inconsistent with substantial selective app opening.

Figure 13: Purchase rates as a function of underlying market conditions when the surge price is artificially restricted to 1.5x



Note: This figure presents the purchase rate by surge generator where the actual price observed by the rider is constrained to be 1.5x.

Tables

Table 1: Summary of Sessions Data

	Full Data	Surge = 1	1 < Surge ≤ 2.0	Surge > 2.0
Surge	1.141	1.000	1.509	2.531
Expected wait time	4.118	4.205	3.731	4.046
Purchase rate	59%	62%	53%	39%
City				
Chicago	22%	20%	29%	32%
Los Angeles	25%	26%	20%	24%
New York	29%	31%	21%	19%
San Francisco	24%	22%	30%	25%
Time of Day				
Evening rush	8%	8%	10%	13%
Morning rush	6%	6%	7%	14%
Slow nighttime	12%	13%	10%	8%
Weekday day	23%	25%	15%	12%
Weekday evening	14%	15%	13%	10%
Weekend day	15%	14%	18%	17%
Weekend evening	6%	6%	7%	6%
Weekend event	15%	14%	20%	20%
Rides in Period				
1 ride in period	6%	6%	4%	4%
1 < rides in period ≤ 3	9%	10%	8%	7%
3 < rides in period ≤ 8	14%	15%	13%	13%
> 8 rides in period	71%	70%	75%	75%
Sessions	47469440	37667052	8135793	1666595

This table presents summary statistics for the analysis sample. Column (1) presents statistics for the entire sample. The remaining columns present statistics for three mutually exclusive and exhaustive subsets of the data: Column (2) presents rider sessions with baseline pricing; Column (3) presents rider sessions facing a moderate surge between 1.0x and 2.0x; and Column (4) presents rider sessions facing a surge greater than 2.0x.

Table 2: Impact of Multiplier on Log Requests

	Log requests			
	(1)	(2)	(3)	(4)
Log multiplier	0.343***	0.321***	0.273***	0.238***
	(0.002)	(0.002)	(0.002)	(0.002)
ETA		-0.059***	-0.061***	-0.061***
		(0.000)	(0.000)	(0.000)
City:				
Los Angeles			-0.093***	-0.094***
			(0.001)	(0.001)
New York City			-0.143***	-0.142***
			(0.001)	(0.001)
San Francisco			-0.252***	-0.261***
			(0.001)	(0.001)
Time of week:				
Morning Rush				-0.044**
				(0.001)
Slow Nighttime				-0.206***
				(0.001)
Weekday day				-0.095***
				(0.001)
Weekday evening				0.055***
				(0.001)
Weekend day				-0.118***
				(0.001)
Weekend evening				0.036***
				(0.001)
Weekend party				0.054***
				(0.001)
Constant	0.580***	0.904***	1.086***	1.181***
	(0.001)	(0.001)	(0.001)	(0.001)
Observations	13,959,454	13,959,454	13,959,454	13,959,454
Adjusted R^2	0.003	0.092	0.108	0.126

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The dependent variable is the log of the requests. Model 1 is the baseline effect of the log multiplier on log requests. Model 2 controls for the estimated time of arrival. Model 3 adds city fixed effects. Model 4 adds controls for the time of week.

Table 3: Observables Measured on Both Sides of the Price Discontinuity

	Post-Discontinuity
Request rate	-0.0201
	(0.0011)
Expected wait time	-0.1293
	(0.0142)
City	
Chicago	0.0073
	(0.0058)
Los Angeles	-0.0078
	(0.0044)
New York	0.0031
	(0.0023)
San Francisco	-0.0023
	(0.0054)
Time of Day	
Evening rush	0.0032
	(0.0037)
Morning rush	0.0031
	(0.0024)
Slow nighttime	-0.0048
	(0.0028)
Weekday day	0.0009
	(0.0033)
Weekday evening	-0.0003
	(0.0003)
Weekend day	0.0028
	(0.0036)
Weekend evening	-0.0012
	(0.0021)
Weekend event	-0.0023
	(0.0052)
Rides in Period	
1 ride in period	0.0007
	(0.0004)
$1 <$ rides in period ≤ 3	-0.0004
	(0.0005)
$3 <$ rides in period ≤ 8	0.0008
	(0.0006)
> 8 rides in period	-0.0012
	(0.0012)

This table presents OLS estimates for differences in key observables across price discontinuities. Standard errors are reported in parentheses.

Table 4: Estimated Price Elasticities at various Points along the Demand Curve

Table 3: Raw and RD Elasticity Estimates

Surge Threshold	(1)	(2)	(3)	(4)	(5)
1.2	-0.26	-0.43	-0.52	-0.52	-0.52
	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)
1.3	-0.32	-0.31	-0.35	-0.36	-0.34
	(0.01)	(0.04)	(0.04)	(0.04)	(0.04)
1.4	-0.42	-0.47	-0.53	-0.53	-0.58
	(0.01)	(0.05)	(0.05)	(0.05)	(0.05)
1.5	-0.42	-0.47	-0.50	-0.50	-0.49
	(0.02)	(0.05)	(0.05)	(0.05)	(0.05)
1.6	-0.33	-0.33	-0.43	-0.45	-0.50
	(0.02)	(0.06)	(0.06)	(0.06)	(0.06)
1.7	-0.62	-0.60	-0.66	-0.68	-0.68
	(0.03)	(0.08)	(0.08)	(0.08)	(0.08)
1.8	-0.73	-0.80	-0.85	-0.88	-0.89
	(0.03)	(0.10)	(0.10)	(0.10)	(0.10)
1.9 - 2.3	-0.77	-0.99	-1.02	-1.06	-1.01
	(0.02)	(0.07)	(0.07)	(0.07)	(0.07)
2.4 - 3.0	-0.37	-0.34	-0.38	-0.39	-0.25
	(0.05)	(0.17)	(0.17)	(0.17)	(0.17)
3.1 - 5.0	-0.72	-0.61	-0.75	-0.78	-0.65
	(0.14)	(0.46)	(0.46)	(0.46)	(0.46)
Source of identification	All variation	RD only	RD only	RD only	RD only
Control for wait time	No	No	Yes	Yes	Yes
Instrument for wait time	No	No	No	Yes	Yes
Additional controls	No	No	No	No	Yes

This table presents price elasticity estimates for each discontinuous jump in price. The surge prices on the left-most column indicate the higher of the two surge prices represented at the threshold e.g. the estimate created using the threshold between 1.2 and 1.3 would be labelled 1.3. Column (1) reports the raw elasticity estimates; Column (2) reports RD elasticity estimates; Column (3) reports RD estimates with expected wait time as a control; Column (4) reports RD estimates where expected wait time serves as an instrument; Column (5) reports instrumented RD estimates with additional controls for city and time of week. Standard errors are reported in parentheses. In each specification, the bottom three rows are the inverse variance weighted estimates taken from multiple discontinuities.

Table 5: Price Elasticities for Various Sub-Populations

	Elasticity
All periods	-0.5463
	(0.0209)
City	
Chicago	-0.6618
	(0.0367)
Los Angeles	-0.3270
	(0.0458)
New York	-0.6084
	(0.0514)
San Francisco	-0.5244
	(0.0378)
Time of Day	
Evening rush	-0.4994
	(0.0659)
Morning rush	-0.5206
	(0.0776)
Slow nighttime	-0.5254
	(0.0675)
Weekday day	-0.4611
	(0.0557)
Weekday evening	-0.5418
	(0.0571)
Weekend day	-0.6615
	(0.0540)
Weekend evening	-0.5388
	(0.0775)
Weekend event	-0.5480
	(0.0433)
Rides in Period	
1 ride in period	-0.4374
	(0.1398)
$1 <$ rides in period ≤ 3	-0.3822
	(0.0914)
$3 <$ rides in period ≤ 8	-0.4603
	(0.0642)
> 8 rides in period	-0.5702
	(0.0229)

This table presents elasticity estimates for sub-populations within the data. The RD regression that generates these estimates instruments for expected wait time and includes all relevant controls; this specification corresponds to that presented in Column (5) of Table 3. Standard errors are reported in parentheses.

Table 6: Comparison of surge with and without positive demand shocks included

	Mean Surge	90th Percentile	Sessions above 2.0	Sessions above 3.0
Full data	1.141	1.5	1,666,595	217,934
Middle demand data	1.070	1.3	290,328	13,334
Low demand data	1.071	1.3	253,931	11,266

This table presents statistics for the full sample and for rides with observed demand around and below predicted demand. Predictions are generated using OLS estimates for the effect of week and hour of week on demand by location. Column (1) presents the mean surge in the given locations; Column (2) presents the surge at the 90th percentile; Column (3) presents the number of sessions with surge > 2.0; Column (4) presents the number of sessions with surge > 3.0.

Table 7: Consumer surplus from UberX under alternative assumptions (2015, $ billions)

Assumption	Section	Four cities	U.S.	Change vs. baseline	CS per $1 spent
Baseline: piecewise elasticities, no WTP above 4.9x	5.2	2.57	6.03	—	1.40
Fifth-order polynomial elasticities	5.3	2.45	5.75	−4.7%	1.33
City-specific elasticities	Fn. 36	2.68	6.29	+4.3%	1.46
Reweighted to base-price observables	5.4	3.16	7.42	+23%	1.72
Selective app opening (approximate upper bound)	5.4	≈2.44	≈5.73	≈−5%	≈1.33
Adjusted for worse outside options at high surge	5.6	2.47–2.49	5.79–5.85	−3% to −4%	1.34–1.36
Highest-surge elasticity extended above 4.9x	Fn. 35	3.44	8.08	+34%	1.88

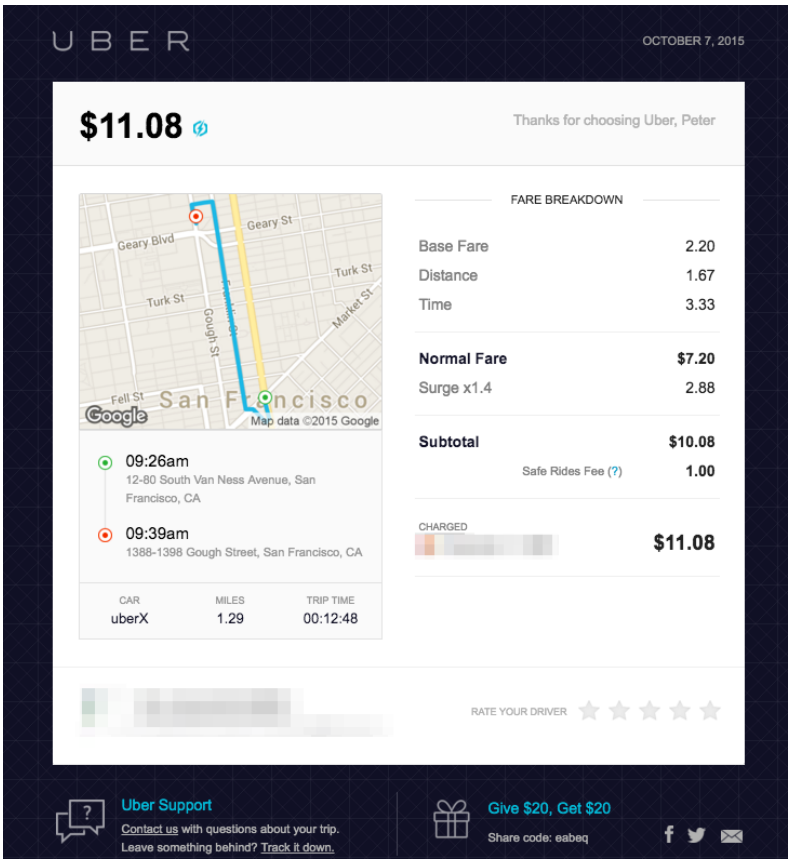
Notes: Figures other than the baseline apply the percentage changes reported in the indicated section to the baseline of $2.57 billion. The selective-app-opening row uses the constant-elasticity approximation described in Section 5.4. Standard errors are not available for the alternative estimates; the baseline bootstrap standard error is $141 million, which likely understates the true uncertainty (Section 4.2). The demand-shock subsamples in Section 5.4 are omitted because they are too noisy at high surge to yield a reliable estimate.

Online Appendices

Time category definitions

Morning Rush = Monday - Friday, between 5:00am and 7:00am
Weekday day = Monday - Friday, between 6:00am and 5:00pm (excluding both rush hours)
Evening Rush = Monday - Friday, between 5:00pm and 7:00pm
Weekday Evening = Monday - Friday, between 7:00pm to 11:00pm
Weekend day = Saturday - Sunday, between 6:00am and 5:00pm
Weekend Evening = Saturday - Sunday, between 6:00pm to 11:00pm
Bar Hours = Thursday 11:00pm to 11:59pm, Friday 12:00am to 3:00am and 11:00pm to 11:59pm, Saturday 12:00am to 3:00am and 11:00pm to 11:59pm, Sunday 12:00am to 3:00am

Example UberX fare



Note: This figure presents the email riders receive upon completion of their ride which details the fare breakdown and additional information about the trip.

First stage expected wait time regression

To estimate the effect of wait times on purchase rates we use the following two specifications, which mirror specifications (4) and (5) in Table 3:

$$Purchase \sim \alpha + \ \beta_1 * Surge + \ \delta_2 * Wait\ Time + \ \varepsilon$$

$$Purchase \sim \alpha + \ \beta_1 * Surge + \ \delta_2 * Wait\ Time + \ \beta_3 * FE(city) + \beta_4 * FE(day\ and\ hour) + \ \varepsilon$$

In both cases, we instrument for both Surge and Wait Time (measured in minutes). The surge instrument is equal to -0.5 when the generator is below a threshold by less than 0.01, 0.5 when the generator is above a threshold by less than 0.01, and 0 otherwise. The wait time instrument follows the same pattern but uses a 3 second window (e.g. if the wait time estimate in seconds was only 2 seconds below the threshold for a higher Wait Time (in minutes), then the instrument would equal -0.5. The coefficient of interest in the first specification, δ_2, is estimated to be -0.0769 (SE = 0.014) and in the second specification $\theta_2$is estimated to be -0.0714 (SE = 0.0040).

In the IV first stage of the first regression, the coefficient on the Wait Time instrument is 0.773 (SE = 0.0143) and that of the second regression specification is 0.769 (SE=0.0140). The correlation of the surge instrument and the eta instrument is 0.0003.

Within surge level estimator specification

Using the full range surges within the data, though sampling according to footnote 40, we estimate the following regression to understand the relationship between the surge generator and purchase rates, holding surge constant:

$$Purchase \ = \ \alpha + \ \theta * generator + \beta_2 * FE(surge) + \varepsilon$$

where **θ** is the coefficient of interest.

Table A1: Observations and R2 from the elasticity regressions in Table 4

	Regression 1		Regression 2		Regression 3		Regression 4		Regression 5	
Price	**Obs**	**R2**	**Obs**	**R2**	**Obs**	**R2**	**Obs**	**R2**	**Obs**	**R2**
1.15	38941618	0	3083118	0.003	3071896	0.01	3071896	0.008	3071896	0.016
1.25	2538255	0	2538255	0	2530723	0.006	2530723	0	2530723	0.008
1.35	2479007	0	2479007	0	2472117	0.005	2472117	0.001	2472117	0.008
1.45	2633854	0	2633854	0	2627641	0.004	2627641	0	2627641	0.008
1.55	2148391	0	2148391	0	2143747	0.004	2143747	0	2143747	0.008
1.65	1392687	0	1392687	0	1389450	0.003	1389450	0	1389450	0.009
1.75	1215467	0	1215467	0	1212134	0.003	1212134	0	1212134	0.009
1.85	1066837	0	1066837	0	1063652	0.002	1063652	0	1063652	0.011
1.95	1018362	0.001	1018362	0	1015516	0.002	1015516	0.001	1015516	0.012
2.05	870317	0	870317	0	867682	0.001	867682	0	867682	0.013
2.15	589228	0	589228	0	587315	0.001	587315	0	587315	0.013
2.25	403273	0	403273	0	401794	0.001	401794	0	401794	0.013
2.35	342356	0	342356	0	340641	0.001	340641	0	340641	0.014
2.45	317191	0	317191	0	315060	0.001	315060	0	315060	0.012
2.55	261044	0	261044	0	258983	0.002	258983	0	258983	0.012
2.65	203789	0	203789	0	201862	0.002	201862	0	201862	0.012
2.75	171293	0	171293	0	169510	0.002	169510	0	169510	0.009
2.85	119604	0	119604	0	118664	0.002	118664	0	118664	0.011
2.95	84740	0	84740	0	84136	0.003	84136	0.001	84136	0.012
3.05	73013	0	73013	0	73013	0	72367	0.001	72367	0.01
3.15	64086	0	64086	0	64086	0	63364	0	63364	0.014
3.25	54447	0	54447	0	54447	0	53582	0	53582	0.015
3.35	46576	0	46576	0	46576	0	45782	0.001	45782	0.012
3.45	43998	0	43998	0	45782	0.005	43439	0.002	43439	0.012
3.55	38620	0	38620	0	43439	0.006	38107	0.001	38107	0.011
3.65	29900	0.001	29900	0	38107	0.005	29415	0.001	29415	0.013
3.75	24833	0	24833	0	24372	0.006	24372	0.002	24372	0.016
3.85	19072	0	19072	0	18688	0.006	18688	0.003	18688	0.017
3.95	14916	0.002	14916	0	14741	0.013	14741	0.001	14741	0.011
4.05	12671	0.002	12671	0	12494	0.012	12494	0.008	12494	0.022
4.15	11205	0	11205	0	10921	0.008	10921	0.009	10921	0.027
4.25	10438	0	10438	0	10246	0.008	10253	0.009	10253	0.024
4.35	7397	0.001	7397	0	7352	0.008	7352	0.015	7352	0.032
4.45	6648	0	6648	0	6392	0.006	6392	0.016	6392	0.028
4.55	6321	0.001	6321	0	5811	0.006	5811	0.01	5811	0.021
4.65										
4.75										

4.85	1778	0.019	1778	0	1771	0.027	1771	0.031	1771	0.019

Table A2: Consumer Surplus Estimates by Surge Price Multiplier

Consumer Surplus	Multiplier
$2,150,004,529.00	1
$54,225,227.49	1.2
$57,968,653.28	1.3
$55,954,123.91	1.4
$113,110,810.30	1.5
$23,770,730.51	1.6
$24,300,213.10	1.7
$15,309,664.18	1.8
$62,430,895.03	2.1
$14,632,524.43	2.7
$0.00	3.9

The table above illustrates how a very large proportion of the CS is driven by trips that are priced at 1.0. That makes sense given that that is simply a large proportion of total trips.